CERTIFICATE

It is certified that the work contained in the project report titled "DESIGN AND IMPLEMENTATION OF AN IOT-ENABLED FLOOD DETECTION AND NOTIFICATION SYSTEM" by VARSHINII SN & (22UEIR0006), ZEBAANJUM N & (22UEID0134), has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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This project report entitled DESIGN AND IMPLEMENTATION OF AN IOT-ENABLED FLOOD DETECTION AND NOTIFICATION SYSTEM by VARSHINII S N (22UEIR0006),ZEBAANJUM N (22UEID0134) is approved for the degree of B.Tech in Computer Science & Engineering.

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BATCH NO:MAP002

DESIGN AND IMPLEMENTATAION OF AN IOT-ENABLED FLOOD DETECTION AND NOTIFICATION SYSTEM

Major project report submitted
in partial fulfillment of the requirement for award of the degree of

Bachelor of Technology
in
Computer Science & Engineering

By

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ABSTRACT

Floods are one of the worst natural disasters because they kill a lot of people and damage a lot of property. Real-time flood monitoring and alert systems are very important for reducing the effects of these kinds of events. This IoT-based flood alert system uses an Arduino Uno, an ultrasonic water level sensor, and a Wi-Fi module to keep an eye on water levels in areas that are likely to flood. The system is made to notice when water levels are rising and send alerts to the right people and residents right away. This way, people can be ready for disasters. The ultrasonic water level sensor (HC-SR04) sends data to the Arduino Uno, which is the main microcontroller. The HC-SR04 accurately measures the distance between the sensor and the water surface, which shows changes in the water level. The system runs on a stable 5V power supply and has a Wi-Fi module (ESP8266) for wireless communication. This lets the system send real-time water level data to a cloud platform or a dedicated server. When the water level goes above a certain level, the system automatically sends alerts through IoT services. These alerts can be sent to mobile devices or web dashboards. This low-cost, effective solution can be used in many places, like riverbanks, dams, and city drainage systems, to help authorities deal with the risks of flooding. IoT technology not only lets you watch things in real time, but it also lets you look at past data to make better predictions and plans for how to deal with flood risks in the future.

Keywords:Cloud-based Monitoring, Disaster Management, Early Warning System, Environmental Monitoring, Flood Monitoring

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LIST OF ACRONYMS AND ABBREVIATIONS

S.NO	ACRONYMS	ABBREVIATIONS
1	CEA	Complex Engineering Activities
2	CEP	Complex Engineering Problem
3	ESP8266	Wi-Fi Microcontroller Module
4	GSM	Global System for Mobile Communications
5	HC-SR04	Ultrasonic Sensor Module
6	IDE	Integrated Development Environment
7	IoT	Internet of Things
8	LCD	Liquid Crystal Display
9	LED	Light Emitting Diode
10	RGB	Red Green Blue (LED Indicator)
11	SDG	Sustainable Development Goals
12	SMS	Short Message Service
13	Wi-Fi	Wireless Fidelity

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Chapter 1

INTRODUCTION

1.1 Introduction

Floods are one of the worst natural disasters because they often kill many people, destroy property, and hurt the economy badly. Flood-prone areas are becoming more likely to experience disasters because of climate change and the fact that extreme weather can happen at any time. To lessen the damage caused by floods, we need real-time monitoring and early warning systems that can send out timely alerts and help us plan ahead for disasters. The Internet of Things (IoT) has emerged as a transformative technology that enables interconnected devices to monitor, collect, and share data in real-time, offering a robust solution to many modern challenges, including disaster management.This IoT-based Flood Alerting System is meant to keep an eye on water levels in places that are likely to flood and send out alerts when they go above certain safety levels. The system includes an Arduino Uno microcontroller, an ultrasonic water level sensor (HC-SR04), and a Wi-Fi module (ESP8266). These parts work together to collect data all the time and send it wirelessly to a cloud platform or a local server.

The core functionality of the system relies on the ultrasonic sensor, which accurately measures the distance between the sensor and the water surface to detect water level changes. The Arduino Uno processes the sensor data and transmits it via the Wi-Fi module, allowing for remote monitoring of water levels in real-time. When the water level surpasses a critical threshold, the system issues alerts through the IoT platform, which may include push notifications to mobile devices or updates to web dashboards accessible by authorities and residents.

This low-cost and easy-to-use solution works in different environments, like riverbanks, dams, and urban drainage systems. It can provide real-time monitoring and early warnings, which are important for reducing flood risks, protecting communities, and minimizing damage. Additionally, the system can store and analyze historical data, which helps with predicting future events and improving

flood preparation and management strategies. In summary, using IoT technology in this flood alert system offers a scalable and effective way to enhance disaster response and community safety in areas at risk of flooding.

1.2 Background

Floods are some of the most common and damaging natural disasters, impacting millions of people around the world each year. They lead to significant harm to infrastructure, homes, farms, and public services, resulting in major economic losses and risks to human life. In many areas, particularly near rivers, dams, and coastlines, floods happen because of heavy rain, overflowing bodies of water, ineffective drainage systems, and climate change. The growing frequency of severe weather events has made monitoring floods and having early warning systems crucial for keeping people safe and lessening the effects of disasters.

Traditional flood monitoring methods often depend on manual observation or few monitoring stations. These methods may not provide timely or reliable information about rising water levels. In many cases, delays in detecting changes in water levels stop authorities from giving early warnings to residents. As a result, communities may not have enough time to evacuate or take preventive measures, which can lead to more loss of life and property. There is a clear need for an automated monitoring system that can continuously track water levels and give immediate alerts when they exceed safe limits.

Recent progress in technology, especially the Internet of Things (IoT), has allowed for the creation of smart monitoring systems. These systems can gather and send environmental data in real time. IoT links sensors, microcontrollers, and communication networks to track physical conditions and transmit the data to cloud platforms or servers. This setup allows both authorities and users to access real-time information remotely and respond swiftly to emergencies. IoT-based monitoring systems are commonly used in areas like smart cities, environmental monitoring, healthcare, and disaster management.

For flood monitoring systems, IoT technology can be applied to sense the water level with sensors and send the data via wireless networks. With the help of sensors like ultrasonic water level sensors, microcontrollers, and Wi-Fi modules, it becomes feasible to monitor the water levels and create automatic notifications when required.

Flood monitoring systems can be implemented in flood-prone locations such as riversides, dams, and city drainage systems.

The application of Internet of Things (IoT) technology to flood detection provides various benefits such as real-time monitoring, remote availability, automatic notifications, and better disaster preparedness. With the adoption of IoT in a flood alerting system, it is possible for the concerned authorities to undertake proactive measures to address any issues that may arise from flooding. Therefore, the design and implementation of an IoT-powered flood detection and notification system can be considered essential in disaster management.

1.3 Objective

This project mainly aims at the development of an IoT based flood monitoring and warning system which can measure the water level in flood-prone locations and notify about any possible flood hazards. Flood is one of the worst forms of natural disasters and any early detection of flooding can save a lot of damage and lives from destruction. In the suggested system, modern day technology like Arduino Uno, HC-SR04 water level sensors, and the ESP8266 Wi-Fi module will be used for monitoring water levels and transmitting the measured data via the Internet.

Yet another goal of this project is to develop an economical and efficient monitoring system that can be installed quickly at any flood-prone area like riversides, dams, drainage system, and even in urban settings. The device is aimed at ensuring data monitoring, remote access, and immediate notification alerts for the concerned stakeholders using IoT platforms and even smart mobile devices. In this way, the process of disaster preparedness is made easier, and immediate actions can be taken in emergency situations.

1.4 Problem Statement

Disaster situations like floods can be detrimental to the lives, infrastructures, and ecosystems of several places across the globe. There are various areas that experience floods where water level monitoring is still done through traditional means like manual observations or monitoring stations that are insufficient for providing accurate results. Conventional systems have been found inefficient and

incapable of providing real-time updates on the water levels that rise gradually. With no real-time monitoring system and alarms in place to notify authorities about any changes, authorities will only get alerted if water levels continue to rise suddenly because of flooding due to excessive rains or overflowing rivers. This will lead to severe losses of life and property.

Another problem with the current flood detection technology systems is the fact that they lack remote accessibility and affordable infrastructure. These systems need expensive devices and complicated installation procedures, which hinder their deployment in rural and remote areas where flood risks are high. Moreover, without real-time communication between the monitoring systems and local authorities, the disaster management becomes less effective. Thus, there is a necessity of developing an efficient system for continuous monitoring of the water level and giving instant alert when the water level goes beyond the limit. It means that a low-cost and fully automatic system capable of monitoring water levels and providing instant notifications is required.

Chapter 2

LITERATURE REVIEW

A.B.J et al., [1] proposed a machine learning-based flood warning system that enhances prediction accuracy using advanced data-driven techniques. Their model utilizes historical and real-time environmental data to predict flood conditions at an early stage. The system focuses on improving decision-making by reducing false alarms and increasing reliability. However, challenges such as data quality, model training complexity, and real-time implementation were identified.

Kiran et al., [2] developed a real-time urban flood forecasting and alert system designed specifically for densely populated areas. Their approach integrates prediction models with alert mechanisms to provide timely warnings. The system improves disaster preparedness and minimizes damage by enabling faster response. It also highlights the importance of integrating urban infrastructure data for better forecasting accuracy.

Kalai Selvi et al., [3] introduced an IoT-based automated water level monitoring and alert system using sensors and cloud platforms. Their system continuously collects and processes real-time data, ensuring accurate monitoring of water levels. It emphasizes scalability and remote monitoring capabilities, making it suitable for large-scale deployment in flood-prone regions.

Gnanaraj et al., [4] designed an IoT-powered flood warning system with improved detection and notification techniques. Their work focuses on integrating multiple sensors with communication modules to enhance system reliability. The system ensures faster alert generation and efficient data transmission, which is crucial during emergency situations.

Rushingabigwi et al., [5] proposed a simulation-based flood forecasting and alerting system. Their approach focuses on analyzing flood scenarios using modeling techniques to improve prediction accuracy. The study highlights the importance of

system design and performance evaluation in understanding flood behavior.

Thakur et al., [6] developed an IoT-enhanced flood monitoring system using intelligent techniques for early warning. Their system incorporates advanced algorithms to improve prediction accuracy and reduce false alarms. It highlights the role of intelligent data processing in enhancing system performance and reliability.

Karthika et al., [7] proposed a solar-powered flood alert system using Arduino, focusing on energy-efficient and sustainable operation. The system ensures continuous monitoring even during power failures, which is critical during disasters. Their work demonstrates the importance of renewable energy integration in IoT-based systems.

Charaan et al.,[8] enhanced IoT-based flood detection using Arduino UNO with a WiFi module. Their system provides a low-cost and practical solution for real-time monitoring and alert generation. It demonstrates efficient communication between hardware and cloud platforms, making it suitable for rural and resource-constrained areas.

Raman and Iqbal [12] developed an IoT-based flood warning system focusing on reliability and timely alert generation. Their system ensures effective disaster management by providing quick notifications and improving communication efficiency during emergencies.

Banupriya et al., [16] proposed a deep learning-based flood and landslide prediction system integrated with IoT technology. Their model uses advanced AI techniques to analyze complex patterns in environmental data. The system improves risk assessment, prediction accuracy, and provides intelligent alerts for disaster management.

2.1 Existing System

There are several flood alerting systems available today based on the Internet of Things that can be utilized to keep track of water levels and send warnings in advance to those in danger. One common type of such systems is the usage of a combination of sensors like ultrasonic and pressure sensors to detect the increase in water level in rivers, lakes, or other bodies of water. This information is then uploaded to a central server using cloud technologies. Once the level of water surpasses certain pre-set points, alerts are sent through SMS or mobile apps to notify relevant officials or citizens about the possibility of floods.

Nevertheless, there are multiple limitations to such systems, which reduce their usefulness to a certain degree. One of the major limitations is the high costs associated with the use and maintenance of such systems. For many communities in underdeveloped regions, such costs may simply be too high and may not allow for an effective utilization of such systems. Additionally, another problem with the use of technology for flooding alerts involves connectivity problems. As many systems require an active connection to the Internet to transfer data from sensors to users, the effectiveness of such systems in remote rural areas is reduced. Moreover, while many systems allow for the collection of real-time information regarding flooding, they are usually lacking sufficient historical data analysis tools, which can help in the prediction of future floodings.

Additionally, some flood alert systems may face issues such as time lag in sending the alerts due to the network being overloaded or faulty communication, thus making the system less efficient when used for early warnings. Another problem that arises with flood alerting systems is maintaining the sensors and the electronic devices that are exposed to factors such as dust, water, and adverse weather conditions. The non-user friendly nature of the system, along with the integration problems with other data visualizing platforms, also pose another challenge.

2.2 Related Work

There have been several research works done on developing efficient flood monitoring and warning systems through the use of advanced technology, including the Internet of Things (IoT), wireless sensor network, and cloud computing.

Researchers have developed flood monitoring and warning systems that make use of sensors for measuring water level and sending the information to the monitoring centers via remote locations. Flood monitoring and warning systems usually rely on sensors like ultrasonic sensors, water level sensors, and pressure sensors for detecting any change in the water level in rivers, dams, and drainage systems.

Some researches have even incorporated the use of cloud computing technology into the IoT-based flood monitoring system, which has resulted in better handling of data storage, analysis, and visualization processes. This type of system enables the officials to track environmental changes from anywhere using web portals or applications on their mobile devices. The continuous evaluation of water level data results in automatic alarms if the water level exceeds a pre-determined level.

Besides the basic monitoring systems, researchers have also considered the incorporation of new-age technologies like artificial intelligence and machine learning for more accurate and better flood predictions. The systems make use of environmental data to determine trends and predict flood scenarios using intelligent data processing methods along with the help of IoT-based sensors.

From the findings obtained from past studies, it can be inferred that IoT technology has played a critical role in designing effective systems for flood monitoring and alerts. Nevertheless, there are certain drawbacks associated with the design and functioning of such systems.

2.3 Research Gap

Despite the development of several IoT-based flood detection systems, there are still some difficulties that exist within the current methods. The first challenge is the costly process involved in the implementation and operation of such systems. The current technologies require sophisticated sensors and elaborate communication systems, thus making it hard to implement them in remote or developing regions where floods are common.

In addition to the problems mentioned above, another drawback of current systems lies in their reliance on stable internet connectivity during the process of data transfer. In flood-prone areas, particularly rural locations, connectivity may not always be guaranteed. This may cause delays in the transfer of water levels data as well as flood notifications.

Further, there is a trend that existing technologies tend to concentrate more on real-time monitoring rather than providing efficient ways of analyzing past data and forecasting events. For instance, without the capability to analyze flood data, it is hard to understand the pattern of floods and make future arrangements accordingly. Additionally, some systems lack friendly interfaces that would allow them to easily analyze data in case of emergencies.

Hence, there arises the need for designing a cost-effective, reliable, and efficient IoT-based flood detection and alert system which would help to keep track of water levels and issue alerts in real-time regardless of the location, which may not have sufficient infrastructural facilities. This will fill the existing gaps in the related literature.

Another significant area where there is still much room for improvement is related to the robustness and reliability of flood monitoring systems in natural environments. Many of these devices are used in laboratory conditions; thus, they do not demonstrate the same efficiency in harsh environmental conditions involving heavy rains, water flows, changes in the water level, etc. Besides, sensors located next to rivers and drainage systems are subject to environmental impacts, which leads to their damage or incorrect measurements. Furthermore, some systems fail to provide appropriate protection against various failures. For example, they may not have backup power supplies or fault tolerance capabilities. Hence, there is a need for a robust system capable of efficient performance under real conditions.

Chapter 3

PROJECT DESCRIPTION

3.1 Existing System

Conventional water level monitoring systems operate with the help of manual monitoring and traditional sensor-based monitoring systems. In many locations, government authorities manually measure the water level in rivers, dams, and reservoirs using conventional water measuring instruments. These water levels are recorded after periodic intervals of time and then analyzed to see whether the water level is near the critical limit. In some instances, basic water measuring devices are used to record the water level, which sends data through communication means such as GSM or wired networks. These monitoring systems notify the concerned authority once the water level surpasses the pre-set threshold level.

Despite this, there are several drawbacks that these IoT-based flood monitoring systems continue to suffer from. One of the main drawbacks of these systems is their relatively high cost, which is attributed to the sophisticated hardware and monitoring tools used in the construction process. Moreover, there is a reliance on steady internet connection for data transmission, which is not always available in certain rural locations. It is also important to note that some of these devices offer only basic monitoring without any advanced analytical functions, which implies that they might lack proper prediction abilities. Consequently, the likelihood of timely warning cannot be guaranteed.

3.2 Proposed System

The proposed system is IoT-enabled Flood Alerting System that will detect water levels in flood-prone zones and alert authorities and residents accordingly. The proposed system comprises an Arduino Uno as the main control system that gathers data from a water level ultrasonic sensor (HC-SR04). The ultrasonic sensor detects

the distance between the water and the sensor and calculates the water level. Once the water level reaches a particular threshold, the Arduino Uno controller analyzes the data gathered by the sensor and transmits the data to the WiFi module (ESP8266). The Wi-Fi module facilitates wireless communication, and the real-time water level is transferred to an IoT server or platform for remote monitoring via the web or mobile applications.

There are various benefits that the recommended system will have over other existing flood monitoring systems. First, the new design system is affordable and straightforward to implement. It is therefore ideal for application in flood-prone and rural settings where high-tech flood monitoring is unaffordable. Secondly, the system is capable of giving instant information and alerts regarding flooding. This way, it becomes easier to make early predictions and respond during emergencies. Thirdly, the IoT technology used in the design will ensure continuous transfer of water level data and remote access. Finally, the system is scalable to multiple deployment sites such as river banks, drainage areas, and dams among others.

Moreover, the design of the proposed system ensures reliability and maintenance easiness. Since the proposed system will consist of widely available and uncomplicated components such as Arduino Uno, ultrasonic sensors, and Wi-Fi modules, the implementation process does not require advanced technical skills. The system will continuously track water levels and automatically transmit any anomalies by sending alerts. As a result, manual tracking of water levels will become unnecessary. Moreover, the system will allow collecting and storing the gathered data either via cloud technology or a local server, which will be useful for the purpose of further analysis and flood prediction. With the help of such data, authorities will gain insight into the water level patterns and develop their disaster management plans accordingly.

3.3 Feasibility Study

A feasibility study is an integral stage that should be undertaken before the development of any project because it evaluates the ability to implement the proposed system using the existing resources. This paper proposes a feasible, cost-effective, and easy-to-deploy system that can be used to detect and notify people about the presence of floods in their surroundings. As seen, the system incorporates

readily available hardware components, including Arduino Uno, ultrasonic water level sensor, and ESP8266 Wi-Fi modules, thereby making the whole process of implementing the system inexpensive. The system monitors the water level and notifies users in case of reaching a certain water level threshold. As the system uses IoT technology, remote access to collected information will be possible since the collected data can be accessed on the web platform or mobile applications. Moreover, the installation of the system around rivers, dams, and drainage systems will not be complicated because of its small size.

3.3.1 Economic Feasibility

The economic feasibility analysis examines the affordability of the proposed system and its potential return on investment. The proposed IoT-enabled system for flooding detection and alerts is expected to be highly cost-efficient compared to conventional systems of the same kind. Most flood monitoring solutions require costly equipment and expensive infrastructure that involves huge costs for installation and maintenance. The proposed system includes cheap hardware such as Arduino Uno, ultrasonic sensor, and ESP8266 components that are widely available in stores at lower prices. Moreover, the system needs very few installation and maintenance expenses, which makes it applicable in rural or developing areas where the budget is limited. Besides being cost-effective, the system can help minimize the impact of floods on the economy because timely alerts can help prevent disasters. For example, flooding alerts will allow protecting human life and property thus preventing any further losses associated with disaster restoration work.

3.3.2 Technical Feasibility

The technical feasibility of the project is the assessment of the technological resources and technical skills needed to build the system. The suggested IoT flood detection and notification system is technically feasible since it consists of already existing technology and hardware parts that can be found and used easily. It utilizes an Arduino Uno microcontroller as the controlling part of the device for the data processing coming from the ultrasonic water level sensor. An ultrasonic sensor is used to measure the distance between the sensor and the water level in order to assess the water level. There is also an ESP8266 Wi-Fi module that helps with data transfer

in wireless conditions. Such technologies are popular among Internet-of-Things solutions, and there are a lot of programming platforms to use, such as Arduino IDE, etc. Moreover, the proposed solution is quite easy to install, operate, and maintain. Since all the technologies are available and reliable, the proposed project is considered to be technically feasible.

3.3.3 Social Feasibility

The social feasibility of the project examines the effect of the proposed system on the society and its acceptance by the users and communities. Floods may result in damage of property and infrastructure, and even loss of lives of people living in the affected area. Hence, it would be greatly advantageous to implement a flood monitoring and prediction system for the benefit of the society. IoT based flood detection and alerting system is useful for the communities because they will get to know about the increasing level of water and take measures before facing any harm from the floods. This technology will also make communities aware about floods that can occur due to natural causes such as heavy rainfall. Moreover, this technology also increases public safety and helps in disaster preparation by continuous monitoring of the level of water. Since the technology is economical and user friendly and helpful for the society, it is expected that there will be a wide acceptance of this technology among authorities and citizens.

3.4 System Specification

The flood prediction and notification system using the Internet of Things (IoT) technology is meant to measure and give notifications about the level of water within the region where flooding is prone to occur. The Arduino microcontroller is the core element of this system. It collects data through sensors and controls the entire operation of the system. The ultrasonic sensor calculates the distance between itself and the water level to calculate the height of the water. Once the height surpasses the pre-set limit, alerts are generated, and the data is sent remotely via the Wi-Fi module.

Components Used

- **Microcontroller:** Arduino Uno used as the main processing unit for reading sensor data and controlling system operations.

- **Water Level Sensor:** Ultrasonic Sensor (HC-SR04) used to measure the distance between the sensor and the water surface to detect water level changes.
- **Communication Module:** ESP8266 Wi-Fi module used for wireless communication and sending real-time data to an IoT platform.
- **Display Unit:** I2C LCD Display used to show water level status and system messages.
- **Alert System:** Buzzer used to provide audible alerts when the water level crosses the predefined threshold.
- **Visual Indicators:** Three-in-one RGB LED used to indicate safe, warning, and danger levels through different colors.
- **Programming Tool:** Arduino IDE used for writing, compiling, and uploading the program to the Arduino board.
- **Power Supply:** 5V DC power supply used to power the Arduino and connected components.

3.4.1 Tools and Technologies Used

Arduino IDE

The Arduino IDE stands for Integrated Development Environment, which is an open-source application for programming, compiling and uploading of code to the Arduino board. The IDE is compatible with programming languages C and C++. It is an easy-to-use environment for building applications for embedded systems. For the current project, the Arduino IDE will be employed in writing the code for reading the sensor input data, processing the water level measurements, and controlling the output actuators like buzzer, LEDs, and LCD display. Moreover, Arduino IDE helps establish a communication channel between the microcontroller and the Wi-Fi module for transmission of live data.

Table 3.1: Component Specifications and Uses in Flood Alerting System

Component	**Uses**	**Specifications**
Arduino Microcontroller	Controls overall system operations and processes sensor data to trigger alerts.	5V operating voltage, ATmega328P, 14 digital I/O pins, 6 analog inputs
Ultrasonic Sensor	Measures distance between sensor and water surface to detect water level.	Range: 2 cm – 400 cm, Accuracy: ±3 mm, Operating voltage: 5V
Wi-Fi Module (ESP8266)	Enables wireless communication and sends alerts to users via internet.	3.3V operation, 802.11 b/g/n, TCP/IP support
I2C LCD Display	Displays real-time water level and system status.	16x2 display, I2C interface, Operating voltage: 5V
Buzzer	Provides audible alert when water level exceeds threshold.	Operating voltage: 3–5V, Sound output: 85 dB
RGB LED	Gives visual indication of water level status using colors.	5V operation, RGB color output, Common anode/cathode

3.4.2 Standards and Policies

Water Level Sensor / Measuring Device

Water level measuring instruments are utilized to assess the amount of water found in a river or any flood-prone location. It is expected that these measuring devices will be precise, durable, and able to function effectively under different environmental factors like temperature, humidity, and atmospheric pressure. The materials selected must be strong enough to withstand corrosion and other environmental elements.

Standard Used: IS 15118 (2002)

Water Level Gauge (Direct Reading Device)

Water level gauges are direct measures that measure water levels directly without the need for any conversion. Water level gauges must be easy to operate, maintain, and reliable. Water level gauges must also be calibrated appropriately and installed at stable locations to obtain accurate measurements. Water level gauges are usually installed close to water bodies or stilling wells.

Standard Used: IS 9116 (2002)

Flow Measurement Guidelines

Flow measuring criteria offer a systemized method of selecting the best method for measuring water flow taking into account such factors as the environment, accuracy, and kind of water bodies involved. This criterion assists in determining whether a particular flow measuring method is applicable in an online water monitoring system like flood detection.

Standard Used: IS 9922 (2010)

The flood warning system is compliant with several international standards to enhance its reliability, quality, and safety. The ISO 9001 standard ensures quality assurance through all phases of development. Standards by the IEC offer recommendations on electronics and environmental testing to guarantee durability and precision of the system. The WiFi communication between devices is guided by IEEE 802.11 standards for Internet connectivity. Integration with cloud computing platforms is covered by the ISO/IEC 27001 standard on information security.

Chapter 4

SYSTEM DESIGN AND METHODOLOGY

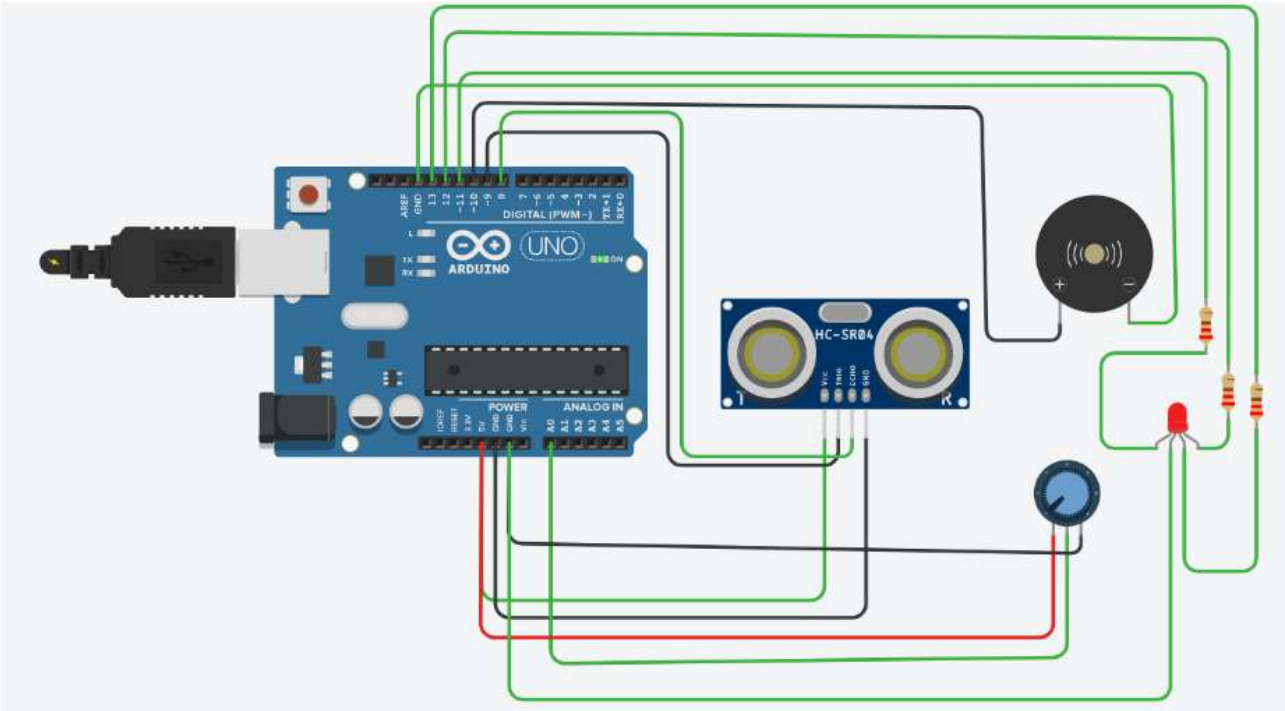

Figure 4.1: **Architecture Diagram of Flood Alerting System**

Description:

Figure 4.1 represents the architecture for the flood alerting system, which is based on IoT, highlighting the relationship between various parts of the system and illustrating their functions within the system. The core of the system is an Arduino microcontroller, which serves as a processing unit to gather data from the ultrasonic sensor (HC-SR04). This ultrasonic sensor calculates the distance from itself to the surface of the water body by emitting high-frequency sound waves and measuring the reflection time, thus calculating the height of water at the monitored site. Then, the Arduino analyzes the collected data and determines if the water level has exceeded the required threshold. The current water level status is shown on an I2C LCD screen, and the buzzer triggers once the critical water level is reached to issue a sound alert.

4.1 Design Phase

4.1.1 Data Flow Diagram

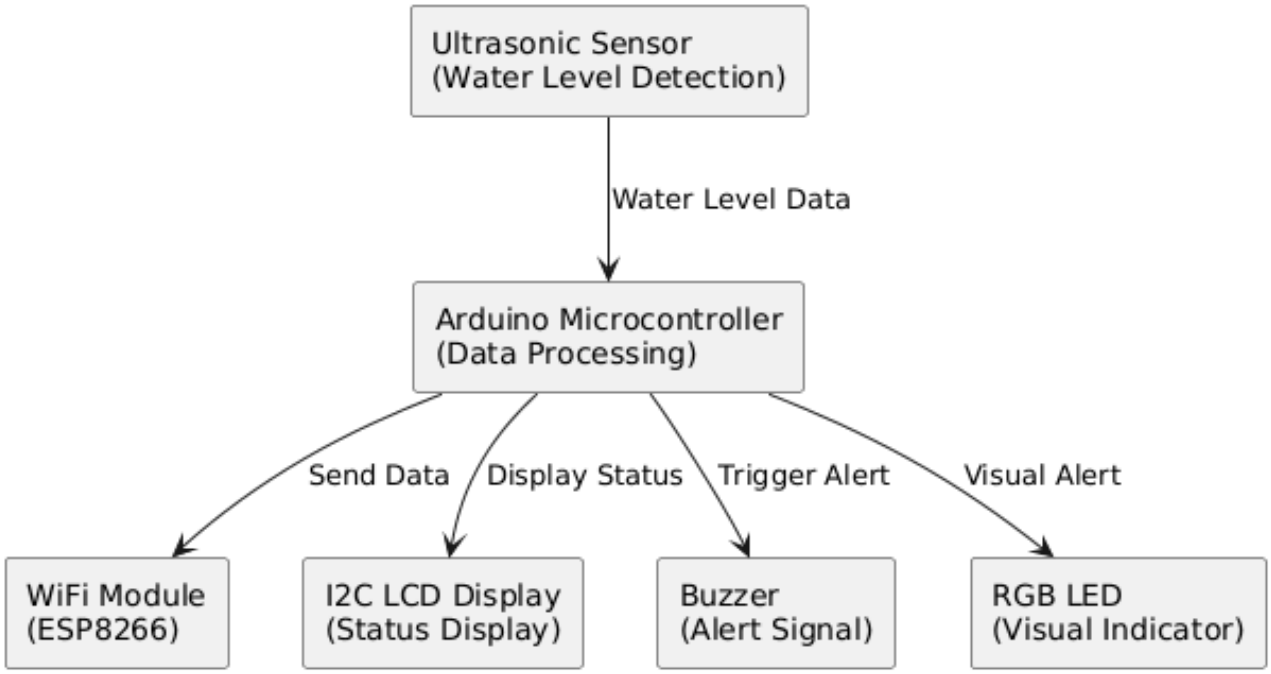

Figure 4.2: **Data Flow Diagram of Flood Alerting System**

Description:

Figure 4.2 represents the data flow diagram of the IoT-based flood alerting system, where the flow of data among different blocks of the system is shown. The working of the system starts with the ultrasonic sensor, as it senses the water level by calculating the distance from the sensor to the water surface. The water level data sensed by the sensor is sent to the Arduino microcontroller that serves as the brain of the entire system. Arduino processes the data and assesses the water level condition accordingly. On the basis of the assessed data, different functions are performed by the Arduino microcontroller. Water level information is communicated using the ESP8266 Wi-Fi module for remote monitoring purposes. In addition, the water level status is indicated on the I2C LCD Display for local monitoring. If the water level is greater than the defined threshold value, Arduino triggers the buzzer to produce an audio alarm and also lights up the RGB LED.

4.1.2 Use Case Diagram

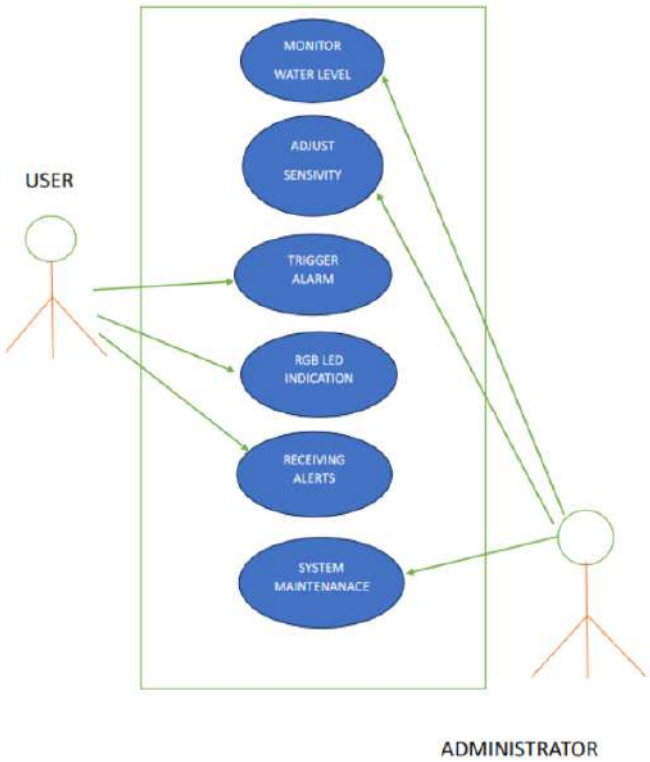

Figure 4.3: **Use Case Diagram of Flood Alerting System**

Description:

Figure 4.3 represents the use case diagram that provides a graphical illustration of the interaction between the main actors, namely the administrator and people in the flood warning system, highlighting their duties. The administrator plays an integral role in the system since he performs vital activities like checking water levels and setting up the system. By observing data through sensors, the administrator is able to react to high water levels; consequently, the community is protected. Moreover, it is also the responsibility of the administrator to issue alerts using both visible means (leds) and audible ways (buzzers). Through these means, the administrator warns the people on any hazards and alerts them to take appropriate measures.

4.1.3 Class Diagram

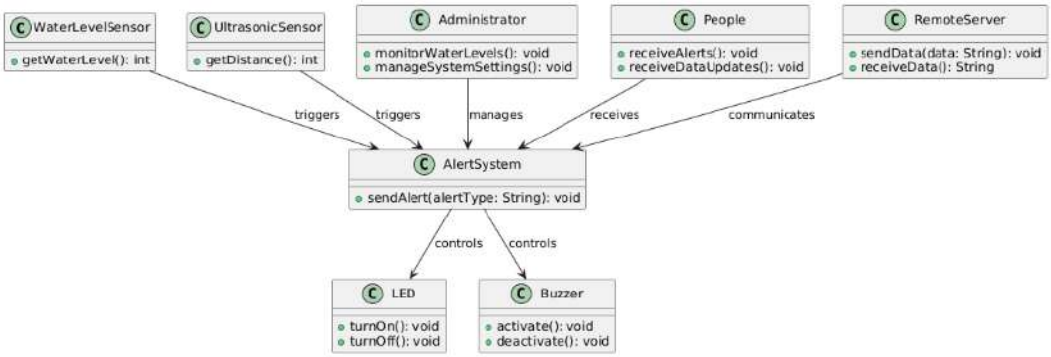

Figure 4.4: **Class Diagram of Flood Alerting System**

Description:

Figure 4.4 represents the class diagram for the flood alert system architecture, which includes the various classes involved and how they interact. The two important classes in the system include the water level sensor and ultrasonic sensor classes, whose primary responsibility is to determine water levels and distances. The alert system class is another crucial part of the system, which triggers alerts when necessary. The alert system class works hand-in-hand with the LED and Buzzer classes, which control the visuals and sounds generated from the alert system. Administrator class also plays a critical role in this system since it controls the system as a whole by regulating water levels and settings. People class is another element of the system, but unlike the previous classes, people class consists of individuals who use the alert system. These users depend on the system to give them updates regarding water levels and any alerts related to floods. Remote server class also plays an integral part in this system by making sure that data exchange takes place effectively between the alert system and its users.

4.1.4 Sequence Diagram

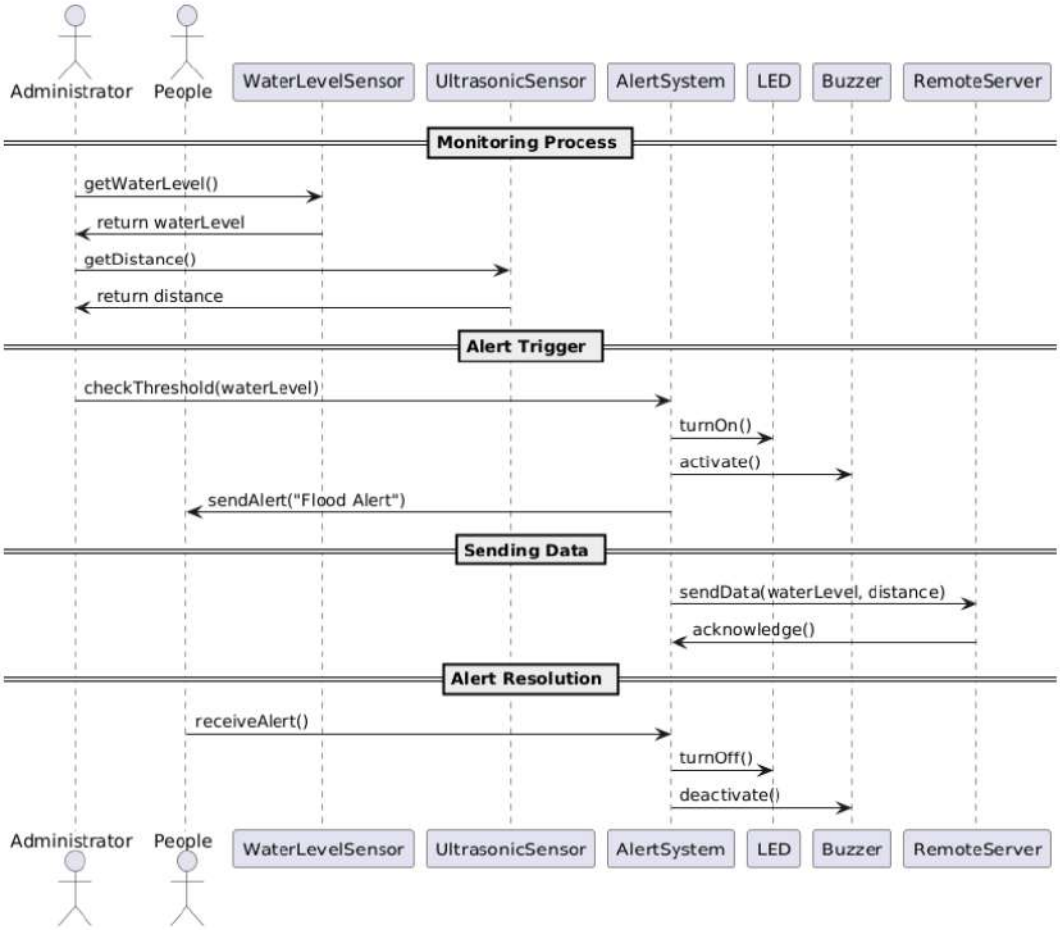

Figure 4.5: **Sequence Diagram of Flood Alerting System**

Description:

Figure 4.5 represents the sequence diagram showing the interaction of the flood alert system in the process of monitoring and raising alerts. This diagram begins with the administrator fetching the information regarding the water level reading from the water level sensor and distance reading from the ultrasonic sensor. Upon detecting the water level above the predefined limit, the alert system is triggered, resulting in the alerting of the People via the activation of the LED and Buzzer. Data transmission is then done to the Remote server and confirmation sent back from the server. After the alert is confirmed by the people, the alerts are switched off, showing how all entities interact to raise alerts and protect against floods.

4.1.5 Collaboration diagram

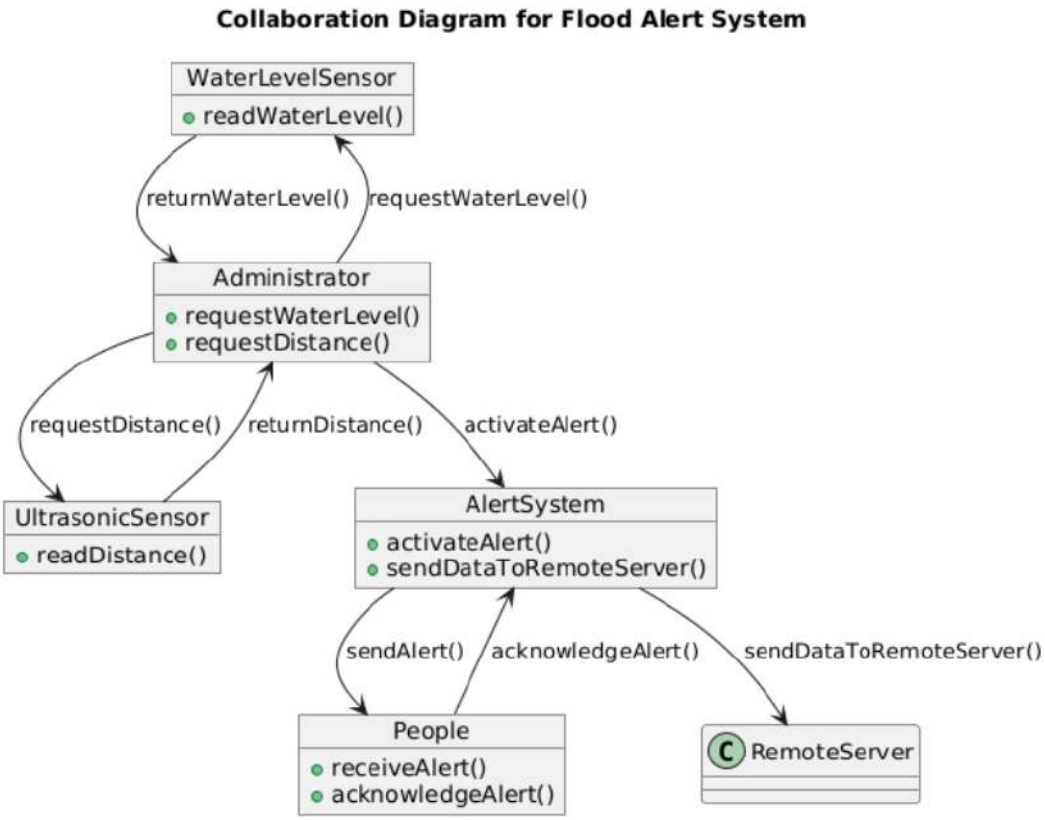

Figure 4.6: **Collaboration diagram of Flood Alerting System**

Description:

Figure 4.6 represents the collaboration diagram, which shows the interaction of the main elements of the flood alert system. It clearly indicates the interaction of the Administrator, the water level sensor, the ultrasonic sensor, the alert system, the People, and the remote server. The administrator sends requests for the levels of water and the distance to be sent by the sensors. In case the level of water rises above a certain threshold, the administrator triggers the alert system. After that, the alert system triggers alerts to the people. The people will acknowledge the received alert. The alert system receives the acknowledgement of the alert from the People. Finally, the alert system sends essential information to the remote server.

4.1.6 Activity Diagram

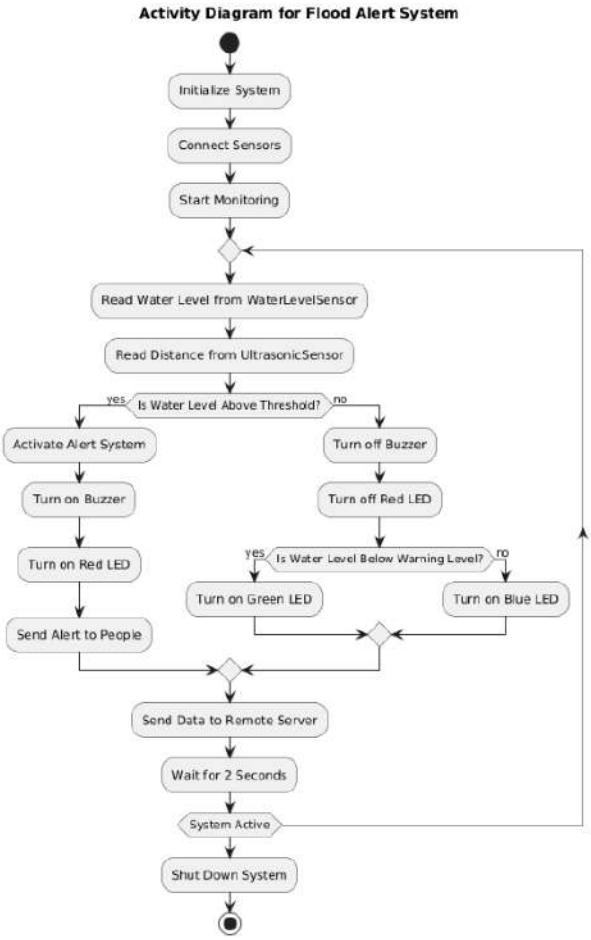

Figure 4.7: **Activity Diagram of Flood Alerting System**

Description:

Figure 4.7 is the representation of the activity diagram of the workflow of the flood alert system, beginning with the initialization of the system and connecting of sensors to monitoring activities. The process starts with the connection of the water level sensor and the ultrasonic sensor, and it continues with a loop where readings of water level and distance are made. Should the water level exceed the preset value, the alarm will be sounded, and the buzzer and the red LED will turn on, along with the alert being sent to people. In case the water level is lower than the warning threshold, then the system will light up the green LED otherwise, it will light up the blue LED. In all cases, after each loop is done, the system will send its readings to the server and wait for two seconds.

4.2 Algorithm & Pseudo Code

4.2.1 Algorithm

Threshold-Based Algorithm:
Description: The algorithm for threshold-based alerts works by constantly monitoring the environmental data, which in this example is the height of water measured by a sensor. The algorithm works in a cycle where it reads the current height of the water. Once the height drops below a certain threshold (10 cm), an alert is triggered.
Key Features:

- Real-time Monitoring: The algorithm creates simulations of the sensor data to determine the water level at all times.
- Alert System: The alert is generated only if the water level is below the threshold value, thus avoiding unnecessary alerts since it will be generated only once, until there is an occurrence of change.
- Reset Mechanism: Once the water level reaches a safe point, the alert status will reset itself.

This method works well as an early warning system in flood-prone areas, allowing for timely action to avoid any damage and hazards

1. First, deploy the sensors and make sure that you are connected to the network, if required.
2. Specify certain threshold levels for every individual sensor (for example, if the distance is less than 50cm then it is high water level, specific light intensity
3. Read sensor data constantly.
4. Interpret sensor readings in order to determine whether the reading is above or below the specified threshold level.
5. When the threshold is breached, act accordingly.
6. Turn on LEDs, activate the buzzers, and transfer the data online via cloud services.
7. Store or send the sensor data for analysis purposes.

8. Repeat this cycle over and over again.

4.2.2 Pseudo Code

```
START

Initialize Arduino Microcontroller
Initialize Ultrasonic Sensor (HC-SR04)
Initialize WiFi Module (ESP8266)
Initialize I2C LCD Display
Initialize RGB LED Indicator
Initialize Buzzer

Set predefined water level thresholds
    SAFE_LEVEL
    WARNING_LEVEL
    DANGER_LEVEL

Display "Flood Monitoring System Started" on LCD

WHILE system is running DO

    Measure distance between ultrasonic sensor and water surface
    Store the measured value as water_level

    IF water_level is greater than SAFE_LEVEL THEN
        Display "Water Level Safe" on LCD
        Turn ON Green LED (Safe indication)
        Turn OFF Yellow LED
        Turn OFF Red LED
        Turn OFF Buzzer
        Send water level data to IoT platform using WiFi module

    ELSE IF water_level is between WARNING_LEVEL and SAFE_LEVEL THEN
        Display "Water Level Warning" on LCD
        Turn ON Yellow LED (Warning indication)
        Turn OFF Green LED
        Turn OFF Red LED
        Turn OFF Buzzer
        Send warning level data to IoT platform using WiFi module

    ELSE IF water_level is between DANGER_LEVEL and WARNING_LEVEL THEN
        Display "Water Level High" on LCD
        Turn ON Red LED
        Turn OFF Green LED
        Turn OFF Yellow LED
        Activate Buzzer for alert
```

```
        Send alert message and water level data to IoT platform

    ELSE IF water_level is less than DANGER_LEVEL THEN
        Display "Flood Alert!" on LCD
        Turn ON Red LED continuously
        Activate Buzzer continuously
        Send emergency alert notification through WiFi module
        Log the water level data in IoT platform

    ELSE
        Display "Sensor Error" on LCD
        Turn OFF all LEDs
        Turn OFF buzzer

    END IF

    Wait for few seconds
    Repeat the monitoring process

END WHILE

STOP
```

4.3 Module Description

The IoT based Flood Alerting System consists of various modules that perform various tasks to ensure efficient monitoring and data processing. Each individual module plays a separate role within the system. The working of all modules is coordinated to monitor the water levels and generate an alert whenever the water level reaches above a pre-defined limit.

4.3.1 Water Level Monitoring Module

The task of monitoring the water level is assigned to the Water Level Monitoring Module, which will measure the current water level in the area being monitored. For detecting the distance between the sensor and the water surface, the Water Level Monitoring Module uses an ultrasonic sensor (HC-SR04), which detects the distance by sending ultrasonic signals to the surface of the water. The ultrasonic signals are

sent to the water surface, and the time taken to receive its reflection back is measured by the sensor. The time difference measured is used to calculate the distance between the sensor and the water surface. This distance is then used to calculate the current water level.

4.3.2 Processing and Communication Module

Processing and Communication Module The Processing and Communication module forms the backbone of the system and consists of the Arduino controller as well as the Wi-Fi module (ESP8266). The Arduino gets data from the ultrasonic sensor and analyzes this data to get information about the water level status. The system uses predetermined thresholds for determining whether the water level is normal, dangerous, or alarming.

The function of the Wi-Fi module is that it helps in communicating data from the system to an IoT portal in real time. Users are then able to access the information on their level of water from remote internet applications. The communication module is used for ensuring quick dissemination of information about any probable flooding cases.

4.3.3 Alert and Display Module

The alert and display Module provides visual and audible notifications to indicate the water level condition. This module consists of an I2C LCD display, RGB LED indicator, and buzzer. The LCD display shows real-time water level information and system status messages. The RGB LED provides visual alerts using different colors to represent various alert levels such as safe, warning, and danger conditions.

The module used for alerts and display provides visual and sound-based alerts. It is made up of an I2C LCD display, an RGB LED, and a buzzer. The LCD display is responsible for displaying the current water level information and any other information concerning the operation of the system. An RGB LED is used as an alert light by providing alerts in different colors.

4.4 Steps to execute/run/implement the project

4.4.1 Step 1: Hardware Setup

The process of putting in place the flood alert system starts by connecting all the necessary hardware and making the appropriate connections.

- Connect the HC-SR04 ultrasonic sensor with the Arduino microcontroller for measuring the water level.
- Link the I2C LCD screen with the Arduino for displaying the system's status and water level.
- Connect the RGB LED to provide visual alerts for safe, warning, and danger states.
- Connect the buzzer with the Arduino for producing the sound alerts.
- Connect the Wi-Fi module (ESP8266) for communicating wirelessly.
- Supply all the hardware modules with 5V of constant power.

4.4.2 Step 2: Programming and Configuration

This involves programming the Arduino microcontroller to be able to process the received information and give output accordingly.

- Install and run the Arduino IDE application to write and upload the code to the microcontroller.
- Program the Arduino board to receive data in the form of distance measured by the ultrasonic sensor.
- Set the threshold levels for the various water levels like safe, warning, and danger.
- Program the LCD screen to output data on the water level of the tank.
- Program the RGB LED light to output the water condition depending on the color.
- Program the buzzer to activate when the water reaches the danger level.

4.4.3 Step 3: System Execution and Testing

Once the hardware installation and coding process is done, the system will be tested to check whether it works as expected.

- Turn on the flood alerting system built using the Arduino board.
- Note down the values provided by the sensors on the LCD display.
- Perform the test for various water levels.
- Check if the colors of the RGB LED are altered depending upon the level of the water.
- See if the buzzer is turned on once the water level exceeds the predetermined value.
- Also check if the Wi-Fi module communicates the data remotely.

Chapter 5

IMPLEMENTATION AND TESTING

5.1 Input and Output

Input of the flood alerting system comprises sensor data and output comprises alarm signals and visualization. The flood alerting system constantly keeps track of water levels and sends appropriate alerts depending on whether water levels exceed set thresholds. An Arduino microcontroller is used for processing input data and controlling output devices.

5.1.1 Input Design

Input design entails the technique used in acquiring data and feeding it into the system for analysis. The main source of input in the proposed flood warning system is acquired from the ultrasonic sensor (HC-SR04). This sensor calculates the distance between itself and the water surface through ultrasonic wave transmission and reception of the reflected echo signal. The Arduino microcontroller uses the data from the ultrasonic sensor to calculate the level of the water. The level of the water is continuously analyzed in relation to predetermined threshold levels to determine its status.

5.1.2 Output Design

Output design involves the design of how the processed information will be displayed to the user. In this project, output is done using visual and audio outputs. The LCD monitor displays the water level status while also displaying messages from the system. This allows users to know the status of the monitored region. Also, the water level status is represented by RGB LEDs which show safe, warning, and dangerous water levels. In case of a dangerous level of water level, the buzzer goes off to produce an audio output signal to the user.

5.2 Testing

The testing process is carried out in order to guarantee the proper working of the flood warning system. The system was subjected to different water levels to check whether the sensors, microcontroller, and warning systems work effectively. This involved checking all the components of the system to see if they perform well when subjected to varying water levels.

5.2.1 Testing Strategies

Several techniques were employed to test the validity of the system. The first technique used is functional testing, which was done to confirm that each device works. For instance, the ultrasonic sensor, Arduino, LCD, buzzer, and RGB LED were tested to make sure they perform their roles effectively. Integration testing was also done to confirm that all the devices work well together to form an integrated system. The second technique is simulated testing where changes were made on the distance between the ultrasonic sensor and the water level.

5.2.2 Performance Evaluation

System evaluation was done using measures such as accuracy, speed of response, and reliability. The ultrasonic sensors used were capable of giving precise water level measurements, ensuring that the system detected variations accurately. The Arduino microcontroller was able to process the water level information from the sensors promptly, providing timely responses. The RGB LED lights and the buzzer functioned perfectly whenever there was a rise in water level beyond the set threshold values. The LCD display was also effective at displaying current water level readings.

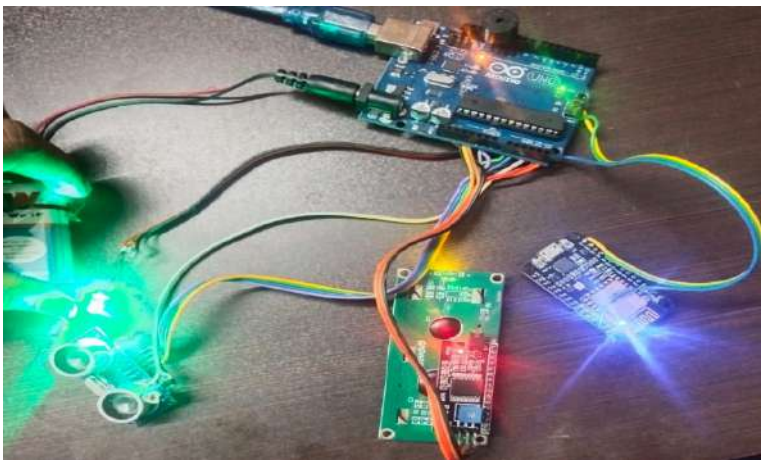

Figure 5.1: **RGB LED Displaying Green Colour Indicating Safe water Level**

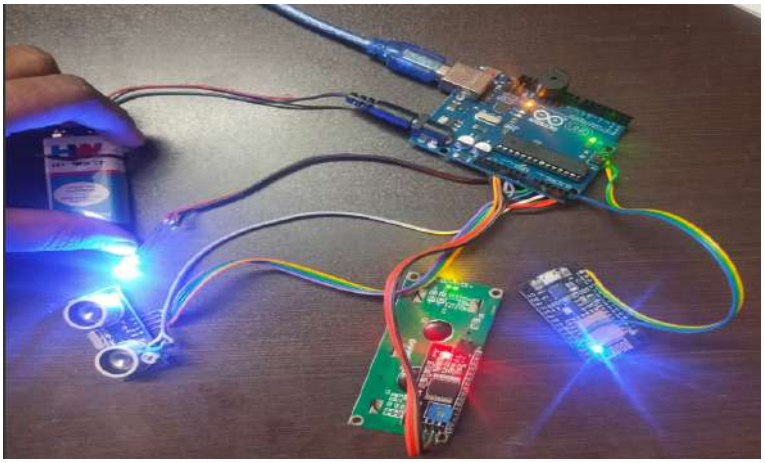

Figure 5.2: **RGB LED Displaying Blue Colour Indicating Precaution Level**

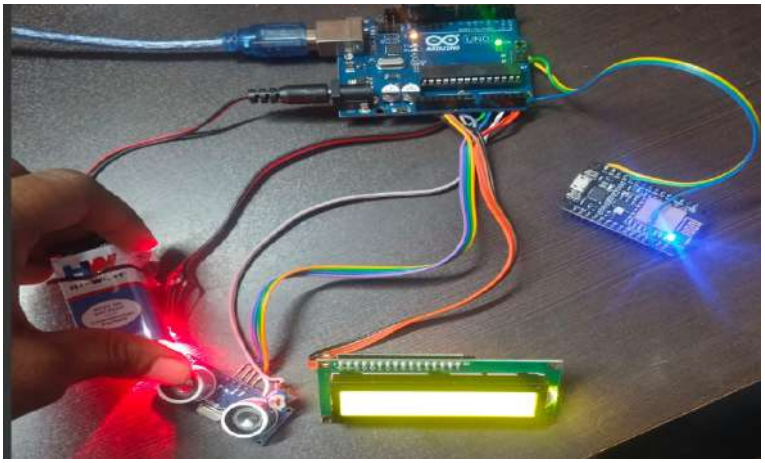

Figure 5.3: **RGB LED Displaying Red Colour Indicating Flood Danger Level**

Chapter 6

RESULTS AND DISCUSSIONS

6.1 Efficiency of the Proposed System

The IoT-Based Flood Alerting System will be developed in order to offer efficient and timely monitoring of the water level in the affected area. The water level will constantly be monitored by the ultrasonic sensor, and the collected data will then be processed by the Arduino microcontroller. The IoT-Based Flood Alerting System compares the water level with its corresponding thresholds and identifies whether the water level is safe, caution, or flood. The implementation of the RGB LED ensures that the water level is indicated by distinct colors depending on the state. If the water level is safe, the color is green; caution is blue, while the flood stage is red.

Another crucial feature of the proposed system is the ability to transfer real-time data using the Wi-Fi module to IoT platforms, which makes remote monitoring of floods possible. The information about water level condition provided by the I2C LCD display helps users to have access to immediate data about the condition of the system. The use of low-cost components such as Arduino, ultrasonic sensors, and Wi-Fi modules allows one to ensure that the costs of the development are minimal. In general, the presented system is highly efficient due to the possibility to detect water level in a fast and accurate manner, generate alerts, respond quickly, and monitor floods remotely. The second advantage of the system under analysis is its ability to transfer real-time data using the Wi-Fi module, thus providing remote monitoring of the condition. The third advantage is related to the fact that the system can be developed using low-cost components such as Arduino and simple electronic modules.

6.2 Comparison of Existing and Proposed System

Existing system:

In the currently used flood monitoring systems, there is no continuous water level monitoring, and the water level measurement is done manually or with help of basic alarm sensors. The traditional methods involve the use of human monitoring of water levels or the use of manual devices installed around the river and dam areas. This method does not offer any continuous monitoring process because it is difficult to monitor water levels constantly since sudden water level changes can hardly be detected using such approaches. In addition, alarms are set when water levels reach dangerous stages; hence, there is insufficient time to take the appropriate actions to manage the disaster. One of the weaknesses of the traditional flood monitoring system is that it cannot allow remote monitoring of the flooding situation at various locations.

Proposed system:

The new system suggested in this report is called IoT based Flood Alerting System. It will help monitor the water level with the help of an ultrasonic sensor. The water level will be measured regularly by the system and then compared with the set thresholds to determine the present condition. As soon as the water level starts increasing, the system will provide alerts using RGB LEDs which will show the safety levels of each color. If the water level exceeds the threshold value, a buzzer alert will be issued for nearby people. Apart from local alerts, the Wi-Fi module helps in transmitting the data in real-time to an IoT system so that remote monitoring can also be done. The advantages of this new system over the existing one include the fast reaction time, automatic monitoring, and improved accuracy.

Chapter 7

INDUSTRY DETAILS

7.1 Industry name

EdgeVerve Systems Limited.

7.1.1 Duration of Internship (From Date - To Date)

The duration of the internship is from 02 February 2026 to 31 July 2026.

7.1.2 Duration of Internship in months

The duration of the internship is six months.

7.1.3 Industry Address

The industry address is EdgeVerve Systems Limited, Plot No. 47, Electronics City, Hosur Road, Bengaluru – 560100, India.

7.2 Internship offer letter

7.2.1 VTU21376 offer letter

edgeverve
An Infosys company

April 2, 2025

PMO/MEM/16259997/2025

Varshinii SN

NO.102/53, C BLOCK,2ND CROSS STREET,, THANIKACHALAM NAGAR,PONNIAMMANMEDU,
Chennai - 600110
Tamil Nadu, India

Dear Varshinii,

Subject: Offer of Employment with EdgeVerve Systems Limited

At "Edge Verve", our purpose is to amplify human potential and create the next opportunity for people, businesses and communities. Our employees are key to realizing this. It is our continuing ambition to find and retain people who believe in our values and share our desire to build a great Company.

Congratulations! Further to your application for employment with us, and the subsequent selection process, we are delighted to offer you employment in the role of **Systems Engineer-EV** with EdgeVerve Systems Limited (the "Company"). Your Role Designation will be **5613BSYSEV : Systems Engineer - EV.** Your location of reporting is **Chennai SEZ, India** and the scheduled date of joining the Company is **August 3, 2026**. You shall be a part of the Finacle division within the Company.

Your **Total Gross Salary** including **Performance Bonus** will be INR **30,000** per month. Kindly refer to the compensation details sheet for details.

The terms & conditions of our offer are as follows:

1. Date of Commencement of Employment:

Your employment with the Company shall commence on **August 3, 2026**.

2. Location:

Your current location of employment shall be **Chennai SEZ**, India. However, please note that the Company reserves the right to transfer its employees to any of its Unit (s), office(s) or of its Affiliates and/or the office of the Company's customer ("Work Location"), other than the one an employee was initially hired to work in.

For the purpose of this agreement, "Affiliate" means any entity that controls, is controlled by, or is under common control with the Company. For purpose of this agreement, "control" means possessing, directly or indirectly, the power to direct or cause the direction of the management, policies or operations of an entity, whether through ownership of voting securities, by contract or otherwise.

Accordingly, we want you to be aware that you may be transferred to another Work Location in the future depending on the Company's business requirements, and in the event thereof your remuneration and other benefits in relation thereto shall be determined in accordance with the Company's existing and future policies in that Work Location. Please be advised that you, by accepting this offer, hereby give your irrevocable consent to the foregoing.

EDGEVERVE SYSTEMS LIMITED
Plot No. 47, Electronics City,
Hosur Road,
Bengaluru 560 100, India
T 91 80 3952 2222
F 91 80 2852 1300

Registered Office:
CIN: U72200KA2014PLC073660
Plot No. 44, Electronics City
Hosur Road
Bengaluru 560 100, India
www.edgeverve.com
edgeverve@edgeverve.com

1

In the event of a transfer of your Work Location, the Company shall do its best to accommodate, solely at its own sole discretion, any requests or preferences you may have with respect to your choice of Work Location or any other matter in relation to the transfer, consistent with internal Company's policies.

3. Probationary Period and Confirmation as a Permanent Employee:

The period of confirmation with the Company is six months, or such extended date as notified by the Company, from the date of joining. In the event you are currently under probation with Infosys Limited and/or its Affiliates, your probation will continue with the Company for the remaining probation period agreed with your previous Infosys group employer. If you are a confirmed employee with Infosys Limited and/or its Affiliates, you will join as a confirmed employee with the Company.

Please note that as per the existing Company policy, employees would be confirmed on the 1st day of the month succeeding the month of completion of probation period. In addition to your performance during the probationary period, your confirmation as a permanent employee is subject to your submitting the requisite documents as required by the Company. Your confirmation or probation extension shall be as per the Confirmation Policy of the Company. Ongoing employment however, is conditional upon your performance in relation to the accountabilities and level of professionalism expected of you in your role.

4. Earned and Sick Leave:

You shall be eligible for 20 working days of earned leave annually.
The earned leave eligibility shall begin in the respective quarter of your joining the Company. Please note that earned leave days shall be credited on a pro-rated basis in the first quarter of your employment.

Year / Quarter	Quarter 1	Quarter 2	Quarter 3	Quarter 4	Total
Number of Days	5	5	5	5	20

You will also be eligible for 5 days of sick leave. The entire sick leave eligibility shall begin at the time of your joining the company and will be credited on your date of joining the Company.

The Company's Policies also provide for Maternity, Paternity and Bereavement Leave. All leaves will be governed by the Company's leave policy. Further details will be provided to you during the course of your employment.

5. Performance Bonus:

The Company's discretionary performance bonus is governed by various parameters viz. performance of the Company, product / platform / function / department and individual performance. The actual payout shall be determined by the Company in line with above parameters. To be eligible for the payment of this performance bonus for a given period, you should be on the rolls of the Company as on the last working day of the period under consideration. For the avoidance of doubt, the payment of and timelines of payment of performance bonus will be governed by Company's communications on performance bonus payouts. Further details will be provided to you during the course of your employment.

6. Superannuation / Superannuation Allowance:

In the event you are eligible for Superannuation / Superannuation Allowance, it shall be calculated at 15% of your Basic Salary as mentioned in the compensation details sheet of this offer. Of the amount as calculated above, a maximum amount of INR. 12,500 per month shall be contributed towards the Superannuation Fund of the Company; the balance amount shall be paid to you on a monthly basis as a part of your monthly salary. The eligibility for Superannuation shall be determined as per Company's policy and practices as notified to you during the course of your employment.

7. Health Insurance Plan - Group Health Insurance Scheme

You will be eligible to participate in the Company's group health insurance scheme, and will be covered under the standard plan of the Company. Further details will be provided to you during the course of your employment.

8. Group Life Insurance & Personnel Accident Coverage Scheme:

You will be eligible to be covered under the Company's group life insurance scheme. A fixed monthly contribution shall be deducted every month to be eligible for coverage under this scheme. Further details will be provided to you during the course of your employment.

9. Notice Period:

During probation, you can terminate your services with one (1) month's written notice or salary thereof. Post confirmation, you can terminate your services with three (3) month's written notice or salary thereof. Similarly, during probation, the Company can terminate your services by giving one (1) months' notice or salary thereof. Post confirmation, the Company can terminate your services by giving three (3) months' notice or salary thereof. However, the Company shall not be required to give any notice period when terminating your services for misconduct. Further, where circumstances make it necessary, the Company will have the discretion to relieve you only at the end of such period which is mutually agreed.

10. Background check:

The Company may, at its discretion conduct background checks prior to or after your expected joining date to validate your employment details furnish by you including your identity, the address provided by you, your education details and details of your prior work experience, if any, and to conduct any criminal checks. You expressly consent to the Company conducting such background checks. In this connection, you expressly consent to the Company migrating employment data submitted by you or available with Infosys Limited or its Affiliates in relation to your employment for the purpose of the Company being able to regulate your employment with the Company. If the Company is not satisfied, in its sole discretion, with the outcome of the background checks, the Company reserves the right to withdraw this offer without notice and compensation or to take any appropriate action against you, including, but not limited to termination of your employment. When a background check raises any concerns regarding any of the details furnished by you and the Company feels the need to further validate such facts, the Company may at its sole discretion, ask you for further information, to substantiate the details that you have earlier provided to the Company, before initiating appropriate action.

11. Other Terms & Conditions:

Our offer to you is subject to the fulfillment of the following conditions:

(a) You, having fully and lawfully completed your graduation / post-graduation qualification, including completion of all applicable educational and examination, requirements for the award of the educational qualification(s) recorded by You in your application for employment with the Company.
(b) You, being declared as passed / qualified by the relevant examination authority. The determination of the adequacy or authenticity of all or any of the proofs will be at the Company's discretion.

(c) Your acceptance and enrolment in the Finacle Certification Program **("Program")** and completion of your obligations towards the Program as mentioned in Annexure V of this offer letter.

Finally, your application and / or employment is subject to the background checks process stated in this agreement. Should the Company ascertain that any of the proofs, documents or statements made by you are untrue or misleading, the Company reserves the right to take appropriate action which may include termination of employment.

In addition, you are required to mandatorily furnish a copy of your passport on the date of joining, unless you have already submitted your passport copy to your previous employer which is either Infosys Limited or its Affiliate in which case the passport copy details available with such companies shall be migrated to the Company's records and you expressly consent to such migration. In the absence of the same, you will be required to undergo a criminal background check, the cost of which will be borne by you. This check will be initiated on the date of your joining or during the course of employment.

You agree not to undertake employment, whether full-time or part-time, as the director / partner / member / employee of any other organization / entity engaged in any form of business activity without the consent of the Company. The consent may be given subject to any terms and conditions that the Company may think fit and may be withdrawn at any time at the discretion of the Company.

You will be required to claim all business related expenses and settle all dues within 30 days of incurring the expenditure. Expense claims, which are received after 30 days from the date of incurring the expenditure, will be deemed to be unauthorized. Also, any unsettled dues post the aforementioned period would be deducted from the salary.

You hereby acknowledge and agree to abide by all internal policies of the Company as notified to you from time to time during the course of your employment. The same shall either be expressly provided to you or they shall be available at the Company's intranet.

You shall be required to sign certain mandatory agreements, including but not limited to the Confidentiality, Intellectual Property Rights, the Code of Business Conduct and Ethics and your employment shall be governed by all the rules and regulations, as amended from time to time, of the Company as applicable to your category of employees. This offer is also conditional upon your acceptance and execution of the Non-Compete Agreement (Annexure I) and Confidentiality, Intellectual Property Rights (Annexure II).

You understand and accept that while participating in any social media forum(s) for either professional or personal reasons you shall act responsibly. Social media should never be used in a way that breaches any of our policies. Without being exhaustive you are prohibited from using social media to:

- post any confidential or proprietary information belonging to the company or of its customers, clients, business partners, suppliers, vendors or other stakeholders
- defame or disparage the company or its employees or Affiliates, customers, clients, business partners, suppliers, vendors or other stakeholders
- Breach any law or ethical standards

Please refer to the Company's Social Media Policy for further details.

Furthermore, the Company has various human resources and administration policies and procedures. The Company reserves the right to vary these policies at any time in its absolute discretion and you are required to abide by all applicable policies of the Company as notified to you from time to time during the course of your employment.

This offer of employment constitutes the entire agreement between you and the Company regarding the terms of your employment and it is the complete, final, and exclusive embodiment of your agreement with regard to this subject matter and supersedes any other promises, warranties, representations or agreements, whether written or oral. It is entered into without reliance on any promise or representation

other than those expressly contained herein, and it cannot be modified or amended except in writing signed by an authorised officer of the Company.

The contents of this offer of employment are confidential in nature and you shall not disclose any information pertaining to your terms of employment for any purpose without the prior written consent of the Company. Your confidentiality obligations created by this offer of employment shall survive termination of this offer of employment.

If any of the terms or conditions of this offer are found to be illegal or unenforceable, such terms shall be treated as severable from the rest of the terms and conditions of this offer and the remaining terms and conditions shall continue in force.

This agreement shall be governed by the laws of India and you hereby agree to the exclusive jurisdiction of the courts in Bengaluru, India.

As a token of your acceptance of this offer, please bring a duly signed duplicate copy of the same on the date of joining.

We welcome you to the EdgeVerve family and wish you a rewarding career over the years to come.

For EdgeVerve Systems Limited,

Rajeshwar Rao
AVP - Senior Director and Head - Talent Acquisition

I, **Varshinii SN,** have read, understood and agree to the terms and conditions as set forth in this offer letter.

Date:

Signature:

edgeverve
An Infosys company

COMPENSATION DETAILS (All figures in INR. per month)	
NAME	**Varshinii SN,**
ROLE	**Systems Engineer-EV**
ROLE DESIGNATION	**5613BSYSEV : Systems Engineer - EV**
1. MONTHLY COMPONENTS	
BASIC SALARY	15,000
BASKET OF ALLOWANCES (HRA, LTA, Medical Allowance, Children's Education Allowance, Transport Allowance, Miscellaneous Allowance)	9,478
BONUS / EX-GRATIA (95% of the eligible amount (20% of Basic) being paid out on a monthly basis)	2,850
MONTHLY GROSS SALARY	**27,328**

2. OTHER COMPONENT	
BONUS / EX-GRATIA - (Balance 5% will be paid out in the end of the financial year after adjusting the advance (95%) paid out on a monthly basis)	150

3. RETIRAL BENEFITS	
PROVIDENT FUND - 12% of Basic	1,800
GRATUITY - 4.81% of Basic*	722
FIXED GROSS SALARY (1+2+3)	**30,000**

4. PERFORMANCE BONUS	**At an indicative Payout of 100%**
PERFORMANCE-BASED BONUS	0
TOTAL GROSS SALARY (Inclusive of Bonus)	**30,000**

All the above benefits are as per Company's policies, which are subject to change from time to time.

*The gratuity amount set out above is an approximation. The eligibility and payment of gratuity shall be in accordance with the applicable provisions of the Payment of Gratuity Act, as amended and supplemented from time to time.

NON COMPETE AGREEMENT (Annexure I)

I,**Varshinii SN** do hereby acknowledge and confirm the following:-

(1) I am accepting employment with EdgeVerve Systems Limited ("Company"). Now, as per the presents below, I agree to the following terms herein, and acknowledge that this is a material condition of my employment with the Company.

(2) I am required, on behalf of the Company, to provide services to, or solicit business from, various clients of the Company for whom I performed services as a Company employee (each such client hereinafter referred to as a "Customer").

(3) In consideration of the above, I agree that for a period of six (6) months following the termination of my employment with the Company for any reason, I will not:

a. accept any offer of employment from any Customer, where I had worked in a professional capacity with that Customer in the twelve (12) months immediately preceding the termination of my employment with the Company;
b. Seek or accept any employment from a Named Competitor of the Company, if my employment with such Named Competitor would involve me having to work with a Customer of the Company with whom I had worked in the twelve (12) months immediately preceding the cessation of employment with the Company.

Non-Compete Clause and Obligations upon Termination

1. During the course of my employment with the Company, I undertake not to enter upon or carry on, either directly or indirectly, any activity which is similar to or in competition with the business or activities carried on by the Company.
2. I agree to return to the Company upon termination of my employment, or upon the Company's request, all tangible forms of confidential information or any other material, together with all copies thereof and extracts there from. I further agree not to make or retain any copies or extracts of any of the foregoing, and will so represent to the Company in writing upon termination of my employment or upon the Company's request.
3. Notification of new Employer: In the event that I leave the employment of EdgeVerve, I hereby grant consent to notification by EdgeVerve to my new employer about my rights and obligations under this Agreement.
4. Non Solicitation of Employees: I agree that for a period of twelve (12) months immediately following the termination of my relationship with EdgeVerve for any reason, whether with or without cause, I shall not either directly or indirectly solicit, induce, recruit or encourage any of EdgeVerve employees to leave their employment, or take away such employees, or attempt to solicit, induce, recruit, encourage or take away employees of EdgeVerve, either for myself or for any other person or entity.
5. Client and Partner Notification: In the event that my employment with the company is terminated (whether by the Company or by me), or my engagement with a particular client or partner is terminated or changed by the Company, I acknowledge and agree that the Company's management shall be solely responsible for notifying any client or partner of the Company of the fact, timing, and details of such termination or change. In addition, in the event that I have decided to terminate my employment with the Company, I agree that I will first notify the Company of my resignation before discussing it with any client or partner of the Company, and will desist from any such discussions until the Company has had a reasonable period of time in which to notify its clients and partners of the termination of my employment.

The named competitors for your stream are as follows:

1. Accenture Limited
2. Cognizant Technology Solutions Corporation
3. International Business Machines Corporation
4. Oracle Corporation
5. SAP AG
6. Tata Consultancy Services Limited
7. Wipro Limited
8. HCL

Please note that this clause applies to your area of work or domain only.

Place: Employee Signature:

Date: Employee Name: **Varshinii SN**

Acknowledged by EdgeVerve Systems Limited

Rajeshwar Rao
AVP - Senior Director and Head - Talent Acquisition

Confidentiality and Intellectual Property Rights (Annexure II)

To,

EdgeVerve Systems Limited,

In consideration of

a) My employment with the Company and

b) The fact that I will have access to information and technical data which is confidential in nature; I agree as follows:

Company Intellectual Property

1. During the course of my employment with the Company, it is likely that I will come into possession of or become familiar with confidential information relating to identity, personnel, customers, vendors, their present and future requirements, pricing, terms and conditions of contracts and transactions between the customers and vendors, patents, trademarks, copyrights, inventions, prototypes, models, drawings, discoveries, concepts, trade secrets, formulas, source code, methods, plans for acquisition or assignment of products, training material, unique development/servicing/commercial methods and models, prospective customer list and systems that relate to the business of the company, its clients, partners or vendors and that are not a matter of public record, specifications, process information and other technical information used in the business and business plans of the Company. I acknowledge and agree that disclosure of such confidential information can cause serious damage, financial or otherwise, to the Company and put the Company at a serious competitive disadvantage. I hereby agree that no part of such confidential information will be disclosed by me to any person without the prior written approval of the Company. I agree to maintain strict confidentiality in respect of all such confidential information both during my employment and subsequent to my employment with the Company.

2. I further agree that all copyright, ownership and intellectual property rights in any work, including any enhancements or derivatives, of any nature carried out by me during the course of my employment with the Company shall vest with and remain with the Company, or such other person / entity as designated by the Company. I agree that I will not be entitled to claim any right, title or interest therein, including moral rights. In the event it should be established that such work does not qualify as a work made for hire, I agree to and do hereby assign to Company all of my right, title, and interest in such work product including, but not limited to, all copyrights, patents, trademarks, designs and other proprietary rights. I further acknowledge and agree that the Company may, in its sole discretion, assign to third parties all such intellectual property rights.

3. I further agree that I shall not disclose to any person all or any part of the intellectual property rights in any work of any nature carried out by me during the course of my employment, belonging to the Company, or such other person / entity as designated by the Company. I agree to maintain strict confidentiality in respect of all such intellectual property rights both during my employment and subsequent to my employment with the Company. All intellectual property and confidential information, deemed Company's sole and exclusive property, unless warranted by work, shall not be removed from the premises without prior written consent, nor shall I make unauthorized copies of such information.

4. Both during the term of this agreement and thereafter, I agree to fully cooperate with Company in the creation, establishment, protection and enforcement of any intellectual property rights that may derive as a result of the services performed by me under the terms of this agreement. This shall include executing, acknowledging, and delivering to the Company all documents or papers that may be requested by the Company to enable the Company or customers of the Company to create, establish, publish or protect said intellectual property rights.

5. If requested by the Company, or upon cessation of my employment with the Company, I agree to promptly return to the Company all materials, writings, equipment, models, mechanisms, and the like obtained from or through the Company, including, but not limited to, all confidential information, all of which I recognize is the sole and exclusive property of the Company or such other person / entity as designated by the Company.

6. I shall immediately inform the Company and I undertake to maintain the confidentiality of all information pertaining to the Company, its intellectual property including methodologies, processes and know how, and its business activities and agree to be bound by any other obligations owed to the Company in respect of third parties.

7. (a) Customer Intellectual Property – During the course of my employment with the Company I may be deputed on specific project/projects of customers. I agree that during such deputation I will fully, punctually and to the satisfaction of the Company and such customers comply with the rules and requirements relating to non-disclosure, confidentiality, security checks and procedures or any other related matters as may be prescribed by such customers.

   (b) I agree to strictly and conscientiously abide by the rules, regulations and security policies related to network security both at the Company and at customer sites. I shall adhere to the specific security measures at customer sites unequivocally and to the satisfaction of the Company and the customer.

8. I represent and warrant to the Company that I am not a party to or otherwise bound by any agreement that may, in any way, restrict my right or ability to enter into this agreement or otherwise be employed by the Company. Additionally, I agree that I will not reveal to the Company, or otherwise utilize in my employment with the Company, any proprietary trade secrets or confidential information of any previous employer.

9. This agreement shall be binding on and shall inure to the benefit of the parties hereto, and their heirs, administrators, successors, and permitted assigns.

10. No waiver by either party of any default shall be deemed as a waiver of any prior or subsequent default of the same or other provisions of this agreement.

11. If any provision hereof is held invalid or unenforceable by a court of competent jurisdiction, such invalidity shall not affect the validity or operation of any other provision, and such invalid provision shall be deemed to be severed from the agreement.

12. This agreement and the rights and obligations hereunder are personal with respect to employee and may not be assigned by any act of employee or by operation of law. The Company shall, however, have the absolute, unfettered right to assign this agreement to a successor in interest to the Company or to the purchaser of any of the assets of the Company.

13. I agree that the Company may initiate appropriate legal action against me for the breach of any of the terms of this agreement, and to recover the costs of such legal action, including all damages and attorney's fees. The Company may also, at its discretion, terminate my employment with the Company.

14. This agreement constitutes the entire agreement, and supersedes all other previous agreements. It can only be modified by an agreement in writing and signed by the parties hereto. Notwithstanding the foregoing, those contractual obligations relating to (a) intellectual property and confidentiality signed by me with the group companies; or (b) client or vendor partners of the group companies, which survive the cessation of my employment with such group companies shall continue for the duration these obligations are enforceable.

Place: Employee Signature:

Date: Employee Name: Varshinii SN

Acknowledged by EdgeVerve Systems Limited

Rajeshwar Rao
AVP - Senior Director and Head - Talent Acquisition

Offer Annexure for India

At the time of joining, you are requested to bring the following original certificates along with two sets of photocopies (A4 size only). These documents are also essential to obtain a Visa when you may be required in the future to go abroad on business/ work. So it would be in your interest to ensure that they are all in place at the time of joining the organization.

1. Offer letter (signed photocopy as a token of acceptance of the offer)
2. Degree / Provisional / Course completion Certificate of all College / University qualifications
3. Mark sheets of all College / University qualifications (all semesters)
4. Certificates & Mark sheets of 10th & 12th classes
5. 5 passport size photographs
6. Relieving letter from all previous employers (in case you have previous employment experience)
7. Passport (if you don't have a passport, kindly apply immediately)
8. PAN Card (PAN details are mandatory at the time of joining)
9. Four wheeler driving license (where available)
10. NSR Card else ITPIN Number
11. Aadhar Card

Under Income Tax laws, disclosure of your Permanent Account Number (PAN) to EdgeVerve Systems Limited is mandatory. Please disclose your PAN to EdgeVerve Systems Limited on the day of joining as this is required for the income tax calculation on your salary Non Availability of PAN No. attracts an Income tax at higher rates.

As per the directions of the Employees Provident Fund Organization ("EPFO") it is mandatory KYC requirement for all employees to link their Aadhaar card number with their provident fund Universal Account Number ("UAN"). Hence, to comply with the obligations under the Employees Provident Funds and Miscellaneous Provisions Act 1952 ("EPF Act") and also for availing the benefits under the EPF Act, you are required to disclose either your Aadhaar card number or the Aadhaar enrollment number as applicable, at the time of joining. Please note that EdgeVerve Systems Ltd. would not be liable for your inability to avail the benefits provided under the EPF Act due to the non-linkage of your Aadhaar card number to your UAN.

ANNEXURE – V
FINACLE CERTIFICATION PROGRAM

1. For the successful completion of the Program, You must fulfill the below requirements:

a) have a minimum of 80% attendance for the in-person sessions conducted by Edgeverve. In case of absence from any of the in-person sessions, You are expected to attend the recorded sessions in order to complete the training,
b) submit all the assignments and capstone projects within the stipulated timelines,
c) complete your mandatory internship with Edgeverve during your 8th semester. In case of non-completion, absence or opting out from the internship, Edgeverve shall have the right to revoke the offer made to you.
d) satisfy the qualification criteria for all the assessment tests within the relevant topics covered in the sessions.

2. Qualification Criteria:

a) The assessments shall have two components:

(i) Objective type assessment – 30% weightage

(ii) Hands-on problem-solving assessment - 70% weightage

b) The assessments shall be conducted by the end of each focus area. To qualify in the respective focus areas, You must attain an average score of at least 65%, which includes both objective and hands-on problem solving components. In the event of failing to meet the minimum average score of 65% in the assessments, Edgeverve shall provide you an individualized coaching, if needed. Furthermore, you will be given two additional opportunities to take the assessment tests at the end of the seventh semester in order to meet the qualification criteria.

Edgeverve reserves the right to revoke the offer if any of the aforementioned criteria are not met by You.

7.2.2 VTU21713 offer letter

March 27, 2025

PMO/MEM/16259417/2025

Zebaanjum N

No. 2A, Lenin Street, Krishnamoorthy Nagar, Kodungaiyur,
Chennai - 600118
Tamil Nadu, India

Dear Zebaanjum,

Subject: Offer of Employment with EdgeVerve Systems Limited

At "Edge Verve", our purpose is to amplify human potential and create the next opportunity for people, businesses and communities. Our employees are key to realizing this. It is our continuing ambition to find and retain people who believe in our values and share our desire to build a great Company.

Congratulations! Further to your application for employment with us, and the subsequent selection process, we are delighted to offer you employment in the role of **Systems Engineer-EV** with EdgeVerve Systems Limited (the "Company"). Your Role Designation will be **5613BSYSEV : Systems Engineer - EV.** Your location of reporting is **Bangalore, India** and the scheduled date of joining the Company is **August 3, 2026**. You shall be a part of the Finacle division within the Company.

Your **Total Gross Salary** including **Performance Bonus** will be INR **30,000** per month. Kindly refer to the compensation details sheet for details.

The terms & conditions of our offer are as follows:

1. Date of Commencement of Employment:

Your employment with the Company shall commence on **August 3, 2026**.

2. Location:

Your current location of employment shall be **Bangalore**, India. However, please note that the Company reserves the right to transfer its employees to any of its Unit (s), office(s) or of its Affiliates and/or the office of the Company's customer ("Work Location"), other than the one an employee was initially hired to work in.

For the purpose of this agreement, "Affiliate" means any entity that controls, is controlled by, or is under common control with the Company. For purpose of this agreement, "control" means possessing, directly or indirectly, the power to direct or cause the direction of the management, policies or operations of an entity, whether through ownership of voting securities, by contract or otherwise.

Accordingly, we want you to be aware that you may be transferred to another Work Location in the future depending on the Company's business requirements, and in the event thereof your remuneration and other benefits in relation thereto shall be determined in accordance with the Company's existing and future policies in that Work Location. Please be advised that you, by accepting this offer, hereby give your irrevocable consent to the foregoing.

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Plot No. 44, Electronics City
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www.edgeverve.com
edgeverve@edgeverve.com

In the event of a transfer of your Work Location, the Company shall do its best to accommodate, solely at its own sole discretion, any requests or preferences you may have with respect to your choice of Work Location or any other matter in relation to the transfer, consistent with internal Company's policies.

3. Probationary Period and Confirmation as a Permanent Employee:

The period of confirmation with the Company is six months, or such extended date as notified by the Company, from the date of joining. In the event you are currently under probation with Infosys Limited and/or its Affiliates, your probation will continue with the Company for the remaining probation period agreed with your previous Infosys group employer. If you are a confirmed employee with Infosys Limited and/or its Affiliates, you will join as a confirmed employee with the Company.

Please note that as per the existing Company policy, employees would be confirmed on the 1st day of the month succeeding the month of completion of probation period. In addition to your performance during the probationary period, your confirmation as a permanent employee is subject to your submitting the requisite documents as required by the Company. Your confirmation or probation extension shall be as per the Confirmation Policy of the Company. Ongoing employment however, is conditional upon your performance in relation to the accountabilities and level of professionalism expected of you in your role.

4. Earned and Sick Leave:

You shall be eligible for 20 working days of earned leave annually.
The earned leave eligibility shall begin in the respective quarter of your joining the Company. Please note that earned leave days shall be credited on a pro-rated basis in the first quarter of your employment.

Year / Quarter	Quarter 1	Quarter 2	Quarter 3	Quarter 4	Total
Number of Days	5	5	5	5	20

You will also be eligible for 5 days of sick leave. The entire sick leave eligibility shall begin at the time of your joining the company and will be credited on your date of joining the Company.

The Company's Policies also provide for Maternity, Paternity and Bereavement Leave. All leaves will be governed by the Company's leave policy. Further details will be provided to you during the course of your employment.

5. Performance Bonus:

The Company's discretionary performance bonus is governed by various parameters viz. performance of the Company, product / platform / function / department and individual performance. The actual payout shall be determined by the Company in line with above parameters. To be eligible for the payment of this performance bonus for a given period, you should be on the rolls of the Company as on the last working day of the period under consideration. For the avoidance of doubt, the payment of and timelines of payment of performance bonus will be governed by Company's communications on performance bonus payouts. Further details will be provided to you during the course of your employment.

6. Superannuation / Superannuation Allowance:

In the event you are eligible for Superannuation / Superannuation Allowance, it shall be calculated at 15% of your Basic Salary as mentioned in the compensation details sheet of this offer. Of the amount as calculated above, a maximum amount of INR. 12,500 per month shall be contributed towards the Superannuation Fund of the Company; the balance amount shall be paid to you on a monthly basis as a part of your monthly salary. The eligibility for Superannuation shall be determined as per Company's policy and practices as notified to you during the course of your employment.

7. Health Insurance Plan - Group Health Insurance Scheme

You will be eligible to participate in the Company's group health insurance scheme, and will be covered under the standard plan of the Company. Further details will be provided to you during the course of your employment.

8. Group Life Insurance & Personnel Accident Coverage Scheme:

You will be eligible to be covered under the Company's group life insurance scheme. A fixed monthly contribution shall be deducted every month to be eligible for coverage under this scheme. Further details will be provided to you during the course of your employment.

9. Notice Period:

During probation, you can terminate your services with one (1) month's written notice or salary thereof. Post confirmation, you can terminate your services with three (3) month's written notice or salary thereof. Similarly, during probation, the Company can terminate your services by giving one (1) months' notice or salary thereof. Post confirmation, the Company can terminate your services by giving three (3) months' notice or salary thereof. However, the Company shall not be required to give any notice period when terminating your services for misconduct. Further, where circumstances make it necessary, the Company will have the discretion to relieve you only at the end of such period which is mutually agreed.

10. Background check:

The Company may, at its discretion conduct background checks prior to or after your expected joining date to validate your employment details furnish by you including your identity, the address provided by you, your education details and details of your prior work experience, if any, and to conduct any criminal checks. You expressly consent to the Company conducting such background checks. In this connection, you expressly consent to the Company migrating employment data submitted by you or available with Infosys Limited or its Affiliates in relation to your employment for the purpose of the Company being able to regulate your employment with the Company. If the Company is not satisfied, in its sole discretion, with the outcome of the background checks, the Company reserves the right to withdraw this offer without notice and compensation or to take any appropriate action against you, including, but not limited to termination of your employment. When a background check raises any concerns regarding any of the details furnished by you and the Company feels the need to further validate such facts, the Company may at its sole discretion, ask you for further information, to substantiate the details that you have earlier provided to the Company, before initiating appropriate action.

11. Other Terms & Conditions:

Our offer to you is subject to the fulfillment of the following conditions:

(a) You, having fully and lawfully completed your graduation / post-graduation qualification, including completion of all applicable educational and examination, requirements for the award of the educational qualification(s) recorded by You in your application for employment with the Company.
(b) You, being declared as passed / qualified by the relevant examination authority. The determination of the adequacy or authenticity of all or any of the proofs will be at the Company's discretion.

(c) Your acceptance and enrolment in the Finacle Certification Program **("Program")** and completion of your obligations towards the Program as mentioned in Annexure V of this offer letter.

Finally, your application and / or employment is subject to the background checks process stated in this agreement. Should the Company ascertain that any of the proofs, documents or statements made by you are untrue or misleading, the Company reserves the right to take appropriate action which may include termination of employment.

In addition, you are required to mandatorily furnish a copy of your passport on the date of joining, unless you have already submitted your passport copy to your previous employer which is either Infosys Limited or its Affiliate in which case the passport copy details available with such companies shall be migrated to the Company's records and you expressly consent to such migration. In the absence of the same, you will be required to undergo a criminal background check, the cost of which will be borne by you. This check will be initiated on the date of your joining or during the course of employment.

You agree not to undertake employment, whether full-time or part-time, as the director / partner / member / employee of any other organization / entity engaged in any form of business activity without the consent of the Company. The consent may be given subject to any terms and conditions that the Company may think fit and may be withdrawn at any time at the discretion of the Company.

You will be required to claim all business related expenses and settle all dues within 30 days of incurring the expenditure. Expense claims, which are received after 30 days from the date of incurring the expenditure, will be deemed to be unauthorized. Also, any unsettled dues post the aforementioned period would be deducted from the salary.

You hereby acknowledge and agree to abide by all internal policies of the Company as notified to you from time to time during the course of your employment. The same shall either be expressly provided to you or they shall be available at the Company's intranet.

You shall be required to sign certain mandatory agreements, including but not limited to the Confidentiality, Intellectual Property Rights, the Code of Business Conduct and Ethics and your employment shall be governed by all the rules and regulations, as amended from time to time, of the Company as applicable to your category of employees. This offer is also conditional upon your acceptance and execution of the Non-Compete Agreement (Annexure I) and Confidentiality, Intellectual Property Rights (Annexure II).

You understand and accept that while participating in any social media forum(s) for either professional or personal reasons you shall act responsibly. Social media should never be used in a way that breaches any of our policies. Without being exhaustive you are prohibited from using social media to:

- post any confidential or proprietary information belonging to the company or of its customers, clients, business partners, suppliers, vendors or other stakeholders
- defame or disparage the company or its employees or Affiliates, customers, clients, business partners, suppliers, vendors or other stakeholders
- Breach any law or ethical standards

Please refer to the Company's Social Media Policy for further details.

Furthermore, the Company has various human resources and administration policies and procedures. The Company reserves the right to vary these policies at any time in its absolute discretion and you are required to abide by all applicable policies of the Company as notified to you from time to time during the course of your employment.

This offer of employment constitutes the entire agreement between you and the Company regarding the terms of your employment and it is the complete, final, and exclusive embodiment of your agreement with regard to this subject matter and supersedes any other promises, warranties, representations or agreements, whether written or oral. It is entered into without reliance on any promise or representation

other than those expressly contained herein, and it cannot be modified or amended except in writing signed by an authorised officer of the Company.

The contents of this offer of employment are confidential in nature and you shall not disclose any information pertaining to your terms of employment for any purpose without the prior written consent of the Company. Your confidentiality obligations created by this offer of employment shall survive termination of this offer of employment.

If any of the terms or conditions of this offer are found to be illegal or unenforceable, such terms shall be treated as severable from the rest of the terms and conditions of this offer and the remaining terms and conditions shall continue in force.

This agreement shall be governed by the laws of India and you hereby agree to the exclusive jurisdiction of the courts in Bengaluru, India.

As a token of your acceptance of this offer, please bring a duly signed duplicate copy of the same on the date of joining.

We welcome you to the EdgeVerve family and wish you a rewarding career over the years to come.

For EdgeVerve Systems Limited,

Rajeshwar Rao
AVP - Senior Director and Head - Talent Acquisition

I, **Zebaanjum N,** have read, understood and agree to the terms and conditions as set forth in this offer letter.

Date:

Signature:

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COMPENSATION DETAILS (All figures in INR. per month)	
NAME	**Zebaanjum N,**
ROLE	**Systems Engineer-EV**
ROLE DESIGNATION	**5613BSYSEV : Systems Engineer - EV**
1. MONTHLY COMPONENTS	
BASIC SALARY	15,000
BASKET OF ALLOWANCES (HRA, LTA, Medical Allowance, Children's Education Allowance, Transport Allowance, Miscellaneous Allowance)	9,478
BONUS / EX-GRATIA (95% of the eligible amount (20% of Basic) being paid out on a monthly basis)	2,850
MONTHLY GROSS SALARY	**27,328**

2. OTHER COMPONENT	
BONUS / EX-GRATIA - (Balance 5% will be paid out in the end of the financial year after adjusting the advance (95%) paid out on a monthly basis)	150

3. RETIRAL BENEFITS	
PROVIDENT FUND - 12% of Basic	1,800
GRATUITY - 4.81% of Basic*	722
FIXED GROSS SALARY (1+2+3)	**30,000**

4. PERFORMANCE BONUS	**At an indicative Payout of 100%**
PERFORMANCE-BASED BONUS	0
TOTAL GROSS SALARY (Inclusive of Bonus)	**30,000**

All the above benefits are as per Company's policies, which are subject to change from time to time.

*The gratuity amount set out above is an approximation. The eligibility and payment of gratuity shall be in accordance with the applicable provisions of the Payment of Gratuity Act, as amended and supplemented from time to time.

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NON COMPETE AGREEMENT (Annexure I)

I,**Zebaanjum N** do hereby acknowledge and confirm the following:-

(1) I am accepting employment with EdgeVerve Systems Limited ("Company"). Now, as per the presents below, I agree to the following terms herein, and acknowledge that this is a material condition of my employment with the Company.

(2) I am required, on behalf of the Company, to provide services to, or solicit business from, various clients of the Company for whom I performed services as a Company employee (each such client hereinafter referred to as a "Customer").

(3) In consideration of the above, I agree that for a period of six (6) months following the termination of my employment with the Company for any reason, I will not:

a. accept any offer of employment from any Customer, where I had worked in a professional capacity with that Customer in the twelve (12) months immediately preceding the termination of my employment with the Company;
b. Seek or accept any employment from a Named Competitor of the Company, if my employment with such Named Competitor would involve me having to work with a Customer of the Company with whom I had worked in the twelve (12) months immediately preceding the cessation of employment with the Company.

Non-Compete Clause and Obligations upon Termination

1. During the course of my employment with the Company, I undertake not to enter upon or carry on, either directly or indirectly, any activity which is similar to or in competition with the business or activities carried on by the Company.
2. I agree to return to the Company upon termination of my employment, or upon the Company's request, all tangible forms of confidential information or any other material, together with all copies thereof and extracts there from. I further agree not to make or retain any copies or extracts of any of the foregoing, and will so represent to the Company in writing upon termination of my employment or upon the Company's request.
3. Notification of new Employer: In the event that I leave the employment of EdgeVerve, I hereby grant consent to notification by EdgeVerve to my new employer about my rights and obligations under this Agreement.
4. Non Solicitation of Employees: I agree that for a period of twelve (12) months immediately following the termination of my relationship with EdgeVerve for any reason, whether with or without cause, I shall not either directly or indirectly solicit, induce, recruit or encourage any of EdgeVerve employees to leave their employment, or take away such employees, or attempt to solicit, induce, recruit, encourage or take away employees of EdgeVerve, either for myself or for any other person or entity.
5. Client and Partner Notification: In the event that my employment with the company is terminated (whether by the Company or by me), or my engagement with a particular client or partner is terminated or changed by the Company, I acknowledge and agree that the Company's management shall be solely responsible for notifying any client or partner of the Company of the fact, timing, and details of such termination or change. In addition, in the event that I have decided to terminate my employment with the Company, I agree that I will first notify the Company of my resignation before discussing it with any client or partner of the Company, and will desist from any such discussions until the Company has had a reasonable period of time in which to notify its clients and partners of the termination of my employment.

The named competitors for your stream are as follows:

1. Accenture Limited
2. Cognizant Technology Solutions Corporation
3. International Business Machines Corporation
4. Oracle Corporation
5. SAP AG
6. Tata Consultancy Services Limited
7. Wipro Limited
8. HCL

Please note that this clause applies to your area of work or domain only.

Place: Employee Signature:

Date: Employee Name: **Zebaanjum N**

Acknowledged by EdgeVerve Systems Limited

Rajeshwar Rao
AVP - Senior Director and Head - Talent Acquisition

Confidentiality and Intellectual Property Rights (Annexure II)

To,

EdgeVerve Systems Limited,

In consideration of

a) My employment with the Company and

b) The fact that I will have access to information and technical data which is confidential in nature; I agree as follows:

Company Intellectual Property

1. During the course of my employment with the Company, it is likely that I will come into possession of or become familiar with confidential information relating to identity, personnel, customers, vendors, their present and future requirements, pricing, terms and conditions of contracts and transactions between the customers and vendors, patents, trademarks, copyrights, inventions, prototypes, models, drawings, discoveries, concepts, trade secrets, formulas, source code, methods, plans for acquisition or assignment of products, training material, unique development/servicing/commercial methods and models, prospective customer list and systems that relate to the business of the company, its clients, partners or vendors and that are not a matter of public record, specifications, process information and other technical information used in the business and business plans of the Company. I acknowledge and agree that disclosure of such confidential information can cause serious damage, financial or otherwise, to the Company and put the Company at a serious competitive disadvantage. I hereby agree that no part of such confidential information will be disclosed by me to any person without the prior written approval of the Company. I agree to maintain strict confidentiality in respect of all such confidential information both during my employment and subsequent to my employment with the Company.

2. I further agree that all copyright, ownership and intellectual property rights in any work, including any enhancements or derivatives, of any nature carried out by me during the course of my employment with the Company shall vest with and remain with the Company, or such other person / entity as designated by the Company. I agree that I will not be entitled to claim any right, title or interest therein, including moral rights. In the event it should be established that such work does not qualify as a work made for hire, I agree to and do hereby assign to Company all of my right, title, and interest in such work product including, but not limited to, all copyrights, patents, trademarks, designs and other proprietary rights. I further acknowledge and agree that the Company may, in its sole discretion, assign to third parties all such intellectual property rights.

3. I further agree that I shall not disclose to any person all or any part of the intellectual property rights in any work of any nature carried out by me during the course of my employment, belonging to the Company, or such other person / entity as designated by the Company. I agree to maintain strict confidentiality in respect of all such intellectual property rights both during my employment and subsequent to my employment with the Company. All intellectual property and confidential information, deemed Company's sole and exclusive property, unless warranted by work, shall not be removed from the premises without prior written consent, nor shall I make unauthorized copies of such information.

4. Both during the term of this agreement and thereafter, I agree to fully cooperate with Company in the creation, establishment, protection and enforcement of any intellectual property rights that may derive as a result of the services performed by me under the terms of this agreement. This shall include executing, acknowledging, and delivering to the Company all documents or papers that may be requested by the Company to enable the Company or customers of the Company to create, establish, publish or protect said intellectual property rights.

5. If requested by the Company, or upon cessation of my employment with the Company, I agree to promptly return to the Company all materials, writings, equipment, models, mechanisms, and the like obtained from or through the Company, including, but not limited to, all confidential information, all of which I recognize is the sole and exclusive property of the Company or such other person / entity as designated by the Company.

6. I shall immediately inform the Company and I undertake to maintain the confidentiality of all information pertaining to the Company, its intellectual property including methodologies, processes and know how, and its business activities and agree to be bound by any other obligations owed to the Company in respect of third parties.

7. (a) Customer Intellectual Property – During the course of my employment with the Company I may be deputed on specific project/projects of customers. I agree that during such deputation I will fully, punctually and to the satisfaction of the Company and such customers comply with the rules and requirements relating to non-disclosure, confidentiality, security checks and procedures or any other related matters as may be prescribed by such customers.

   (b) I agree to strictly and conscientiously abide by the rules, regulations and security policies related to network security both at the Company and at customer sites. I shall adhere to the specific security measures at customer sites unequivocally and to the satisfaction of the Company and the customer.

8. I represent and warrant to the Company that I am not a party to or otherwise bound by any agreement that may, in any way, restrict my right or ability to enter into this agreement or otherwise be employed by the Company. Additionally, I agree that I will not reveal to the Company, or otherwise utilize in my employment with the Company, any proprietary trade secrets or confidential information of any previous employer.

9. This agreement shall be binding on and shall inure to the benefit of the parties hereto, and their heirs, administrators, successors, and permitted assigns.

10. No waiver by either party of any default shall be deemed as a waiver of any prior or subsequent default of the same or other provisions of this agreement.

11. If any provision hereof is held invalid or unenforceable by a court of competent jurisdiction, such invalidity shall not affect the validity or operation of any other provision, and such invalid provision shall be deemed to be severed from the agreement.

12. This agreement and the rights and obligations hereunder are personal with respect to employee and may not be assigned by any act of employee or by operation of law. The Company shall, however, have the absolute, unfettered right to assign this agreement to a successor in interest to the Company or to the purchaser of any of the assets of the Company.

13. I agree that the Company may initiate appropriate legal action against me for the breach of any of the terms of this agreement, and to recover the costs of such legal action, including all damages and attorney's fees. The Company may also, at its discretion, terminate my employment with the Company.

14. This agreement constitutes the entire agreement, and supersedes all other previous agreements. It can only be modified by an agreement in writing and signed by the parties hereto. Notwithstanding the foregoing, those contractual obligations relating to (a) intellectual property and confidentiality signed by me with the group companies; or (b) client or vendor partners of the group companies, which survive the cessation of my employment with such group companies shall continue for the duration these obligations are enforceable.

Place: Employee Signature:

Date: Employee Name: Zebaanjum N

Acknowledged by EdgeVerve Systems Limited

Rajeshwar Rao
AVP - Senior Director and Head - Talent Acquisition

Offer Annexure for India

At the time of joining, you are requested to bring the following original certificates along with two sets of photocopies (A4 size only). These documents are also essential to obtain a Visa when you may be required in the future to go abroad on business/ work. So it would be in your interest to ensure that they are all in place at the time of joining the organization.

1. Offer letter (signed photocopy as a token of acceptance of the offer)
2. Degree / Provisional / Course completion Certificate of all College / University qualifications
3. Mark sheets of all College / University qualifications (all semesters)
4. Certificates & Mark sheets of 10th & 12th classes
5. 5 passport size photographs
6. Relieving letter from all previous employers (in case you have previous employment experience)
7. Passport (if you don't have a passport, kindly apply immediately)
8. PAN Card (PAN details are mandatory at the time of joining)
9. Four wheeler driving license (where available)
10. NSR Card else ITPIN Number
11. Aadhar Card

Under Income Tax laws, disclosure of your Permanent Account Number (PAN) to EdgeVerve Systems Limited is mandatory. Please disclose your PAN to EdgeVerve Systems Limited on the day of joining as this is required for the income tax calculation on your salary Non Availability of PAN No. attracts an Income tax at higher rates.

As per the directions of the Employees Provident Fund Organization ("EPFO") it is mandatory KYC requirement for all employees to link their Aadhaar card number with their provident fund Universal Account Number ("UAN"). Hence, to comply with the obligations under the Employees Provident Funds and Miscellaneous Provisions Act 1952 ("EPF Act") and also for availing the benefits under the EPF Act, you are required to disclose either your Aadhaar card number or the Aadhaar enrollment number as applicable, at the time of joining. Please note that EdgeVerve Systems Ltd. would not be liable for your inability to avail the benefits provided under the EPF Act due to the non-linkage of your Aadhaar card number to your UAN.

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6. Superannuation / Superannuation Allowance:

In the event you are eligible for Superannuation / Superannuation Allowance, it shall be calculated at 15% of your Basic Salary as mentioned in the compensation details sheet of this offer. Of the amount as calculated above, a maximum amount of INR. 12,500 per month shall be contributed towards the Superannuation Fund of the Company; the balance amount shall be paid to you on a monthly basis as a part of your monthly salary. The eligibility for Superannuation shall be determined as per Company's policy and practices as notified to you during the course of your employment.

7. Health Insurance Plan - Group Health Insurance Scheme

You will be eligible to participate in the Company's group health insurance scheme, and will be covered under the standard plan of the Company. Further details will be provided to you during the course of your employment.

8. Group Life Insurance & Personnel Accident Coverage Scheme:

You will be eligible to be covered under the Company's group life insurance scheme. A fixed monthly contribution shall be deducted every month to be eligible for coverage under this scheme. Further details will be provided to you during the course of your employment.

9. Notice Period:

During probation, you can terminate your services with one (1) month's written notice or salary thereof. Post confirmation, you can terminate your services with three (3) month's written notice or salary thereof. Similarly, during probation, the Company can terminate your services by giving one (1) months' notice or salary thereof. Post confirmation, the Company can terminate your services by giving three (3) months' notice or salary thereof. However, the Company shall not be required to give any notice period when terminating your services for misconduct. Further, where circumstances make it necessary, the Company will have the discretion to relieve you only at the end of such period which is mutually agreed.

10. Background check:

The Company may, at its discretion conduct background checks prior to or after your expected joining date to validate your employment details furnish by you including your identity, the address provided by you, your education details and details of your prior work experience, if any, and to conduct any criminal checks. You expressly consent to the Company conducting such background checks. In this connection, you expressly consent to the Company migrating employment data submitted by you or available with Infosys Limited or its Affiliates in relation to your employment for the purpose of the Company being able to regulate your employment with the Company. If the Company is not satisfied, in its sole discretion, with the outcome of the background checks, the Company reserves the right to withdraw this offer without notice and compensation or to take any appropriate action against you, including, but not limited to termination of your employment. When a background check raises any concerns regarding any of the details furnished by you and the Company feels the need to further validate such facts, the Company may at its sole discretion, ask you for further information, to substantiate the details that you have earlier provided to the Company, before initiating appropriate action.

11. Other Terms & Conditions:

Our offer to you is subject to the fulfillment of the following conditions:

(a) You, having fully and lawfully completed your graduation / post-graduation qualification, including completion of all applicable educational and examination, requirements for the award of the educational qualification(s) recorded by You in your application for employment with the Company.
(b) You, being declared as passed / qualified by the relevant examination authority. The determination of the adequacy or authenticity of all or any of the proofs will be at the Company's discretion.

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7.3 Internship Completion certificate

The internship completion certificate is not available as the internship is still ongoing and ends on July 31st, 2026.

Chapter 8

CONCLUSION AND FUTURE ENHANCEMENTS

8.1 Summary

The IoT-Based Flood Detection and Alert System developed in this project provides a reliable and efficient solution for real-time monitoring of water levels in flood-prone areas. The system is designed using a water level sensor interfaced with an Arduino microcontroller, which continuously collects and processes environmental data. Based on predefined threshold values, the system classifies water levels into safe, warning, and danger conditions. When the water level exceeds the critical limit, the system automatically activates local alerts through a buzzer and LED indicators while simultaneously sending instant notifications to users via a Wi-Fi module integrated with a Telegram bot. This ensures rapid communication and enables timely preventive measures to reduce the impact of flooding.

The project demonstrates the effective integration of embedded systems and IoT technologies for disaster management applications. It emphasises low-cost implementation, ease of installation, and scalability, making it suitable for deployment in both rural and urban environments. The system reduces dependency on manual monitoring and enhances response efficiency by providing continuous and automated alerts. During testing, the system showed consistent performance in detecting water level variations and triggering alerts accurately. However, certain limitations such as dependency on stable internet connectivity and sensor accuracy under extreme environmental conditions, were observed. ,Future enhancements can include integrating multiple sensors for wider area coverage, cloud-based data storage for historical analysis, and machine learning models for predictive flood forecasting. Overall, this project highlights the potential of smart technologies in improving public safety, minimising damage, and reducing disaster risk.

8.2 Limitations

The proposed IoT-Based Flood Detection and Alert System, while effective for basic monitoring, has certain limitations that may affect its performance in real-world conditions. One of the major challenges is its dependence on stable internet connectivity for sending alerts through the Wi-Fi module. In remote or flood-affected areas, network disruptions can delay or prevent notifications from reaching users. Additionally, the system relies heavily on the accuracy of the water level sensor, which can be influenced by environmental factors such as debris, mud, turbulence, and temperature variations. These conditions may lead to inconsistent readings and reduce the reliability of the alerts.

Another limitation is related to the power supply, as the system requires a continuous power source for uninterrupted operation. During extreme weather conditions, power outages are common, which can interrupt monitoring and alert generation. The system also uses fixed threshold values to determine alert levels, which may not be suitable for all environmental conditions or seasonal variations. This lack of adaptability can result in false alarms or missed warnings, reducing the overall effectiveness of the system.

The current design is limited to localized monitoring using a single sensor, which restricts its ability to cover larger geographical areas. For real-world deployment, multiple sensors would be required to monitor different locations simultaneously. Additionally, the system does not include data storage or historical analysis, making it difficult to track long-term water level trends or improve future predictions. The absence of such features limits the system's capability to provide advanced insights for flood management.

Furthermore, the system lacks advanced functionalities such as predictive analysis and intelligent decision-making, which are essential for modern disaster management systems. There is also no redundancy or backup mechanism, meaning that failure of a single component can affect the entire system. Security aspects such as data protection and unauthorized access prevention are not addressed, which could pose risks in connected environments. Enhancing the system with scalable architecture, secure communication, and intelligent analytics would significantly improve its reliability and practical applicability.

8.3 Future Enhancements

The current IoT-Based Flood Detection and Alert System can be significantly improved by expanding its monitoring capabilities and scalability. One major enhancement is the deployment of multiple interconnected sensor nodes across larger geographical areas such as rivers, reservoirs, and urban flood zones. This would enable real-time distributed monitoring and provide more accurate and comprehensive flood data. Integration with cloud platforms can allow continuous data storage, advanced visualization dashboards, and remote access for authorities and users. Additionally, incorporating GPS modules will help in identifying the exact location of rising water levels, improving emergency response and coordination.

Another important improvement is the implementation of machine learning and predictive analytics to analyze historical and real-time data for forecasting flood events. This would transform the system from a reactive alert system into a proactive decision-support system. The development of a dedicated mobile application can enhance user interaction by providing real-time notifications, alerts, and system status updates in a user-friendly interface. The system can also be integrated with SMS-based alert mechanisms to ensure communication even in low internet connectivity areas.

Further enhancements can include the use of solar-powered energy systems to ensure uninterrupted operation during power failures, which are common during extreme weather conditions. The integration of advanced sensors such as pressure sensors, flow sensors, and rain gauges can improve accuracy and provide multi-parameter monitoring. Adding data encryption and security mechanisms will ensure safe transmission of critical data. Moreover, incorporating redundancy mechanisms and backup systems can improve reliability and fault tolerance of the system. Finally, the system can be extended by integrating with government disaster management platforms and early warning systems to support large-scale deployment and coordinated response. The use of AI-based anomaly detection can help identify unusual patterns and potential system failures. Future research can also focus on developing low-power wide-area network (LPWAN) based communication such as LoRa for long-range data transmission. Overall, these enhancements will improve the system's scalability, accuracy, reliability, and effectiveness in real-world flood management scenarios.

Chapter 9

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

9.1 Mapping of the Project to Complex Engineering Problem (CEP)

Table 9.1: Mapping of Project to Complex Engineering Problem (CEP)

Level	**Attribute**	**Characteristics**	**Targeted WKs**	**Justification**
WP1	Depth of Knowledge	Cannot be resolved without in-depth engineering knowledge at the level of K3, K4, K5, K6 or K8	WK3, WK4, WK5, WK6, WK8	Embedded systems, IoT and sensor knowledge.
WP2	Range of conflicting requirements	Involve wide-ranging and/or conflicting technical and non-technical issues	WK4, WK5, WK6, WK8	Balances cost, performance and efficiency trade-offs
WP3	Depth of analysis required	Have no obvious solution and require originality in analysis	WK3, WK4, WK5, WK8	Requires calibration, analysis and optimization
WP4	Familiarity of issues	Involve infrequently encountered or novel problems	WK4, WK8	Handles environmental and real-world challenges
WP5	Extent of applicable codes	Address problems not encompassed by standards and codes of practice	WK6, WK8	No standards; relies on engineering decisions
WP6	Stakeholder involvement and conflicting requirements	Involve collaboration across disciplines and diverse stakeholders	WK5, WK6, WK7	Meets needs of users and authorities.
WP7	Interdependence	Address high-level problems requiring a systems approach	WK3, WK5, WK6	Integrates multiple hardware and software modules

9.2 Mapping of the Project to Complex Engineering Activities (CEA)

Table 9.2: Mapping of Project to Complex Engineering Activities (CEA)

EA No.	**Attribute**	**Characteristics**	**Justification based on the Project**
EA1	Range of Resources	Diverse resources and technologies	The project utilizes large-scale datasets, distributed infrastructure, blockchain platforms, AI frameworks, cloud resources, and skilled human roles such as admins, users, and auditors.
EA2	Level of Interactions	Optimization of complex interactions	Coordinates sensing, processing, alerts, communication.
EA3	Innovation	Creative and research-based solutions	The project innovatively integrates IoT technology with flood monitoring using a multi-level alert system (Green–Safe, Blue–Warning, Red–Danger). It combines both local physical alerts and internet-based notifications to improve disaster preparedness in a cost-effective manner.
EA4	Consequences to Society and Environment	Significant societal impact	Flood detection systems directly impact public safety, disaster management, and infrastructure protection. Failure to provide early warning may result in loss of life and property. The system contributes to reducing disaster risk and improving environmental resilience.
EA5	Familiarity	Extends beyond prior experience	The project applies embedded systems, IoT integration, real-time monitoring, and environmental sensing beyond routine academic experiments. It addresses real-world flood scenarios requiring practical validation and field-based understanding.

9.3 SDG Mapping (CEP Section)

Table 9.3: SDG Mapping for the Project

SDG No.	**SDG Title**	**Relevance to the Project**
SDG 9	Industry, Innovation and Infrastructure	Promotes smart IoT-based disaster monitoring infrastructure.
SDG 16	Peace, Justice and Strong Institutions	Enhances disaster preparedness in urban flood-prone regions
SDG 11	Sustainable Cities and Communities	Supports adaptation and mitigation of climate-related disasters.
SDG 3	Good Health and Well-being	Reduces risk of injury and loss of life during floods.

Chapter 10

PLAGIARISM REPORT

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ORIGINALITY REPORT

9%	5%	2%	4%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

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Figure 10.1: **Plagiarism Report**

Chapter 11

SOURCE CODE

11.1 Alert and Notification

```
#ifdef ESP32
  #include <WiFi.h>
#else
  #include <ESP8266WiFi.h>
#endif
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>
#include <ArduinoJson.h>

// Replace with your network credentials
const char* ssid = "PROJECT";
const char* password = "123456789";

// Initialize Telegram BOT
#define BOTtoken "7836118741:AAG_fjCCT__96OR1QahSYHXM_Er8Q9XOXKo"

// Use @myidbot to find out the chat ID of an individual or a group
// Also note that you need to click "start" on a bot before it can
// message you
#define CHAT_ID "1305160217"

#ifdef ESP8266
  X509List cert(TELEGRAM_CERTIFICATE_ROOT);
#endif

WiFiClientSecure client;
UniversalTelegramBot bot(BOTtoken, client);

// Checks for new messages every 1 second.
int botRequestDelay = 1000;
unsigned long lastTimeBotRan;

const int ledPin = 2;
bool ledState = LOW;

// Handle what happens when you receive new messages
```

```
//void handleNewMessages(int numNewMessages) {
//   Serial.println("handleNewMessages");
//   Serial.println(String(numNewMessages));
//
//   for (int i=0; i<numNewMessages; i++) {
//     // Chat id of the requester
//     String chat_id = String(bot.messages[i].chat_id);
//     if (chat_id != CHAT_ID){
//       bot.sendMessage(chat_id, "Unauthorized user", "");
//       continue;
//     }
//
//     // Print the received message
//     String text = bot.messages[i].text;
//     Serial.println(text);
//
//     String from_name = bot.messages[i].from_name;
//
//     if (text == "/start") {
//       String welcome = "Welcome, " + from_name + ".\n";
//       welcome += "Use the following commands to control your outputs.\n\n";
//       welcome += "/led_on to turn GPIO ON \n";
//       welcome += "/led_off to turn GPIO OFF \n";
//       welcome += "/state to request current GPIO state \n";
//       bot.sendMessage(chat_id, welcome, "");
//     }
//
//     if (text == "/led_on") {
//       bot.sendMessage(chat_id, "LED state set to ON", "");
//       ledState = HIGH;
//       digitalWrite(ledPin, ledState);
//     }
//
//     if (text == "/led_off") {
//       bot.sendMessage(chat_id, "LED state set to OFF", "");
//       ledState = LOW;
//       digitalWrite(ledPin, ledState);
//     }
//
//     if (text == "/state") {
//       if (digitalRead(ledPin)){
//         bot.sendMessage(chat_id, "LED is ON", "");
//       }
//       else{
//         bot.sendMessage(chat_id, "LED is OFF", "");
//       }
//     }
//   }
```

```
//}

void setup() {
  Serial.begin(115200);

  #ifdef ESP8266
    configTime(0, 0, "pool.ntp.org");      // get UTC time via NTP
    client.setTrustAnchors(&cert); // Add root certificate for api.telegram.org
  #endif

  pinMode(ledPin, OUTPUT);
  digitalWrite(ledPin, ledState);

  // Connect to Wi-Fi
  WiFi.mode(WIFI_STA);
  WiFi.begin(ssid, password);
  #ifdef ESP32
    client.setCACert(TELEGRAM_CERTIFICATE_ROOT); // Add root certificate for api.telegram.org
  #endif
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.println("Connecting to WiFi..");
  }
  // Print ESP32 Local IP Address
  Serial.println(WiFi.localIP());
}

void loop() {
//  if (millis() > lastTimeBotRan + botRequestDelay)  {
//    int numNewMessages = bot.getUpdates(bot.last_message_received + 1);
//
//    while(numNewMessages) {
//      Serial.println("got response");
//      handleNewMessages(numNewMessages);
//      numNewMessages = bot.getUpdates(bot.last_message_received + 1);
//    }
//    lastTimeBotRan = millis();
//  }
  if(digitalRead(D1)==HIGH)
  {
     bot.sendMessage(CHAT_ID, "DAM IS GOING TO OPEN", "");
     delay(2000);
  }
}
```

11.2 Arduino

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27,16,2);
const int pingPin = A0;
const int echoPin = A1;
long duration, inches, cm;
#define TRIGGER_PIN  A0  // Arduino pin tied to trigger pin on the ultrasonic sensor.
#define ECHO_PIN     A1  // Arduino pin tied to echo pin on the ultrasonic sensor.
#define MAX_DISTANCE 400
 // NewPing setup of pins and maximum distance.

void setup() {
  lcd.init();
  lcd.backlight();
  Serial.begin(115200); // Open serial monitor at 115200 baud to see ping results.
  pinMode(11,OUTPUT);
  pinMode(A2,OUTPUT);
  pinMode(8,OUTPUT);
  pinMode(9,OUTPUT);
  pinMode(10,OUTPUT);
}

void loop() {
  lcd.clear();
  lcd.setCursor(3,0);
  lcd.print("DAM IS FULL");
  lcd.setCursor(2,1);
  lcd.print("  NORMAL");
  delay(50);
  pinMode(pingPin, OUTPUT);
  digitalWrite(pingPin, LOW);
  delayMicroseconds(2);
  digitalWrite(pingPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(pingPin, LOW);
  pinMode(echoPin, INPUT);
  duration = pulseIn(echoPin, HIGH);
  inches = microsecondsToInches(duration);
  cm = microsecondsToCentimeters(duration);
  Serial.println(cm);
  if(cm>40)
  {
    digitalWrite(9,LOW);
    digitalWrite(8,HIGH);
    digitalWrite(10,HIGH);
  }
```

```
  if(cm>30&&cm<40)
  {
    digitalWrite(10,LOW);
    digitalWrite(9,HIGH);
    digitalWrite(8,HIGH);
  }
  if(cm<20&cm>0)
  {
    digitalWrite(8,LOW);
    digitalWrite(9,HIGH);
    digitalWrite(10,HIGH);
    Serial.println("DAM IS FULL");
    lcd.clear();
    lcd.setCursor(3,0);
    lcd.print("DAM IS FULL");
    lcd.setCursor(2,1);
    lcd.print("  ALERT!!!");
    digitalWrite(11,HIGH);
    digitalWrite(A2,HIGH);
    delay(1000);
    digitalWrite(A2,LOW);
    digitalWrite(11,LOW);
  }
}
long microsecondsToInches(long microseconds) {
   return microseconds / 74 / 2;
}

long microsecondsToCentimeters(long microseconds) {
   return microseconds / 29 / 2;
}
```

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Design and Implementation of An IoT-Enabled Flood Detection and Notification System

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***Abstract*—The increased risk of flooding in certain areas makes it necessary to use flood monitoring and alerting systems, especially during December and near major water bodies. This paper provides a real-time flood alerting system using IoT to give exact accuracy and a measurement of water level data for early flood detection. The developed system always provides an effective solution for flood warnings, helping the people who are in danger. Despite the advantages, implementing the system comes with challenges like high costs and regular maintenance. Using the Wi-Fi module, the system will send notifications through the Telegram app if the water level exceeds a particular level. The proposed system was functionally validated in a real-time setup by performing a water level detection under various circumstances.**

Keywords: IoT, Flood Alerting System, Arduino, Ultrasonic Sensor, Wi-Fi Module, Real-Time Monitoring, Telegram Notification, Disaster Management, Early Warning System, Environmental Monitoring.

I. Introduction

The IoT-Based Flood Alerting System is designed as an IoT Enabled Flood Detection and Notification System, which is based on Internet of Things(IoT) technology and uses Arduino and a Wi-Fi module. In India, monsoon rainfall occurs from June to October. Mostly in Chennai, heavy rainfall occurs between October and December.

Flooding causes severe environmental damage, including loss of human life, infrastructure damage and disruption of natural ecosystems. Geographically, it will affect the type of soil, environmental changes, modification of the ecosystem, and precipitation can play a part in flooding. In 2018, Kerala was declared a disaster-affected region after it was hit by massive monsoon rains.

The loss of human life and damage to the natural ecosystem are devastating consequences of flooding. This leads to infrastructure damage and human factors. Factors such as the volume and speed of the rain, geology, and moisture levels play a major role in determining the flood.

Due to the increased frequency and intensity of floods, climate change leads to loss of life, property, and infrastructure. To overcome this issue, flood alerting systems are required. The blending of an IoT with a flood monitoring system leads to an innovation or development idea for enabling real-time monitoring and data collection from various parameters. The combination of wireless sensor networks with IoT technology directs the transmission of data to a server for evaluation, and the alerts will be sent to the people who are at risk.

Direct Human Monitoring by Local Authorities, Municipality Workers, or community workers will often monitor the water level in rivers and drainage systems. Upon detecting a risk of flooding, higher authorities are informed to take precautions early and initiate the alerting process. Government or Meteorological Departments utilise satellite images, sensors, and weather models to detect or predict the rainfall and declare the warnings. These warnings are communicated to the people through the radio, television, and some mobile SMS messages.

Ultrasonic Sensors detect the water levels and send the information to the Arduino. Then Arduino will check whether the water level is in the correct stage or has reached the threshold value. If the water level reaches the threshold value, an Arduino will send the data to the Wi-Fi module. Then Wi-fi module send the alert messages to the people. Simultaneously the buzzer will make a sound to alert the people, and the LED glows red to indicate danger.

The objective of the proposed work is all about to design an IoT-based flood alerting system and to detect flood risk using threshold values with timely alerts through visual,auddible and mobile-based notifications.

The primary challenge addressed in this research leads to the development of a Flood Alerting System with low-cost, real time monitors and alerts. Based on the limitations identified in the existing system, this work focuses on real-time monitoring, low cost and simplicity. By installing the units in each area leads to monitoring the water level and sends alert notifications to the particular area by offering particular localised accuracy.

II. Literature review

In the development of this project, several IEEE papers were reviewed, and the key data points are briefly explained below:

The paper Flood and Landslide Prediction presents a disaster management system integrated with machine learning, which is used to give real-time alerts.A Flask-Web application and GSM module are used to enable the SMS notification by giving alerts and audio alarms. Additionally, a CNN model is used to review the ground-level images to check the water level and to detect risks like water overflow.

Recently, an IoT-Based Flood Monitoring System was developed in Thailand. The system includes a wireless network to measure the level of water and send the data using GPRS to the server, which gives real-time data. The study concludes that modern sensors can track the real-time water level to send alerts to the people.

Another study presents an IoT-Based Automated Flood Warning System using a Microcontroller and ESP8266. The system focuses on checking the water level, temperature, and humidity to determine whether a flood will occur or not. Sensors measure the water level, and the data is handled by a Microcontroller. When the water reaches a particular value, the alerts are indicated through an LED that glows red and a buzzer makes alarm sounds. This prototype uses the ThingSpeak IoT platform to send real-time alerts and send warnings to a particular area.

III. Related work

Many flood alerting system rely on devices such as radar, satellites, and sensors, and these devices are overpriced and sophisticated. The latest improvement in IoT technology offers a good option with good potential and substitutes. Previous research has demonstrated the application of IoT for environmental monitoring, but few studies have applied it to flood alerting systems specifically. This study expands on earlier research by blending IoT principles with Arduino to create a cost-effective and scalable solution.

A study by Al-Fuqaha et al.(2018) introduced a flood detection system using a wireless sensor network (WSNs) to provide real-time updates. The system uses an Ultrasonic sensor to measure the water level, and the data will be processed by the Arduino as a Microcontroller. Data from the Ultrasonic Sensors was sent to the communication devices like GSM and Wi-Fi without wires. The system successfully explained that it will alert the people and authorities when the water reaches beyond a particular level through mobile devices, which helps to reduce disaster consequences.

Future Developments in Technology: The integration of advanced IoT technology with the Flood Alerting System enables the measurement of water level, soil moisture, and pressure. Integrating the Machine Learning algorithms with the Flood Alerting System can improve the accuracy by using the historical patterns and environmental data to accurately predict data.

The proposed system depends on fixed threshold-based water level detection to ensure simplicity, low cost and real time response. Predictive flood models need data sets and machine-learning algorithms, which leads to an increase in system cost and complexity. Therefore, the present system depends on real-time detection and immediate alert. The combination of predictive and machine-learning algorithms is considered a future enhancement.

IV. System Design

To address the challenges, the architecture of the proposed system integrates with the IoT technology and the hardware components to provide real-time flood alerts through a buzzer and a display in an I2C circuit.

A. Architecture overview

As shown in Figure 1, the system framework consists of water-level sensors, an Arduino microcontroller, and a Wi-Fi module. The water-level sensors measure the water levels, and the Arduino receives the water level data to analyse the risk of flooding. The Wi-Fi module sends the message to the authorities and people at risk.

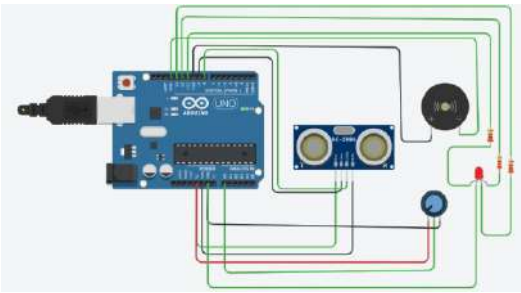

Fig. 1. Architecture Diagram

B. Hardware Components

The flood alerting system consists of several hardware devices, including a Microcontroller, Water-level sensors, Water pressure sensors, a Wi-Fi module, and components required for power supply. Some key features include:

- Real-time flood detection.
- Automated Data sharing.
- Accurate sensor readings.
- Localised Accuracy.

Key functions of Hardware components:

1) Arduino Microcontroller : Collects readings from the sensors and checks whether the value has reached the threshold level. If the threshold is reached, then it will send the information to the Wi-Fi to trigger the alert notifications.

2) Ultrasonic Sensors (HC-SR04): Measure the water levels and transmit the readings to the Arduino.

3) Wi-Fi module : Receives data from the Arduino and sends the alerts and data to the users via the internet using Telegram app.

4) Buzzer: Provides sound alerts when the water level reaches a higher level.

5) LED : Lights up when the water level reaches a higher level.

- Red Light: Danger.
- Blue Light: Time to take precautions.
- Green Light: Safe.

C. Software Components

The Flood Monitoring System and its Notification System are programmed, monitored, and controlled using the Arduino IDE and the Wi-Fi module.

Arduino IDE: Used for programming the microcontroller and configuring sensor data processing.

Cloud Platforms: Stores sensor data and provides a web interface for monitoring and management.

D. IoT Integration

Data collected by the Sensors is transmitted to a cloud platform via the Arduino. The platform analyses data to determine flood risk and triggers alerts based on predefined thresholds.

V. IMPLEMENTATION

A. Water Level Sensor :

This module is used to measure the distance from the surface of water and is a non-contact sensor. Functions:

- The ultrasonic sensor sends the waves to the surface of the water.
- It measures the time taken by the waves to return and collects the data.
- Calculating a distance based on how fast the waves travel.
- Finally, the sensor outputs the measured water level.

B. Arduino Microcontroller:

In the Flood Alerting System, the sensor is attached to the Arduino Microcontroller. The sensor sends the waves to the water surface, then the Arduino will compare the level to a limit. If the level reaches the limit, then the buzzer gives a sound alarm, and the LED will glow up The sensor's wireless technique plays a role in ensuring spot-on and consistent water level readings, making these elements effective components of a flood alerting system. To execute the Flood alerting system, the ultrasonic water level sensor is linked to the Arduino's input pins. Then the Arduino Microcontroller reads the data that was sent by the water level sensor and calculates the water level. There will be a particular level, and Arduino compares the calculated water level to the threshold value. If the water level reaches a particular level, then the buzzer triggers the sound alarm, which leads to the LED lights up. Additionally, the alert will be sent to the mobile by alert messages through the Wi-fi module.

The Arduino Microcontroller works as the brain of the Flood Alerting System, which will read sensor data and execute the code in real time. It also controls the functions of the Buzzer, LED, and Wi-Fi module. The Arduino monitors the functions and fixes some errors to view an output. From these steps, we get to know that the Arduino Microcontroller provides timely alerts and real-time monitoring.

C. Wi-Fi Module :

The Wi-Fi module is used to enable internet connectivity to send an alert to the mobile devices when there is a risk of flooding. Its functions include:

- Enabling Wi-Fi connections between the system components and the access point.
- Sending or receiving the data to or from a server or mobile application.
- Staying connected: Non-stop internet connection.

In the Flood Alerting System, the Wi-fi module connects to the router or access point. The Arduino shares water level data, collected from the water level sensor. When the water level reaches a higher point, the Wi-Fi module sends the alert message to the mobile app, allowing users to receive alerts and monitor remotely.

D. Buzzer :

The Buzzer is an alerting device that produces a sound when the water level reaches the maximum level. The primary function of the buzzer is to alert people in danger. The Buzzer is connected to the Arduino UNO's digital output pins. When the water level reaches the threshold level, then Arduino sends the signal to the buzzer, which leads to triggering an alert.

E. LED :

The LED [Light Emitting Diode] is a visual signal that activates lights to glow when the water level reaches a predefined level. The LED provides a signal alert in case of any chance of flood events. The LED is connected to the

Arduino microcontroller, which sends a signal to activate the light when the threshold level is met.

F. Breadboard and Jumper Wires :

In a Flood Alerting System, Breadboard and Jumper wires are used to interconnect the various components and to make a final design. By using a Breadboard, the components can be placed and blended without the need for soldering. Jumper wires will connect the components to the Arduino, which activates the flow of signals.

G. Power Supply :

The Power Supply offers power to the Flood Alerting System. The role of the power supply is to give constant power even when the power is out. The battery is attached to the Arduino Uno and other components, which provide power to the system. The battery is connected to the Arduino Uno and other components, providing power to the system. The battery can be replaced as needed.

VI. DESIGN METHODOLOGY

Floods are natural disasters causing severe impacts that lead to life loss and damage, especially for the people who suffer a lot during floods. To overcome this issue, it is important to have a real-time monitoring system to alert the people and authorities. The suggested Flood alerting system uses components such as an Arduino, Wi-Fi modules, and IoT technology to monitor the water levels and environmental conditions.

TABLE I
COMPONENT POWER CONSUMPTION

Component	Power Source
Arduino Uno	0.05 W (active)
Wi-Fi Module (ESP8266)	0.10 W (active)
Ultrasonic Sensor	0.05 W (active)
Buzzer	5V Regulated Power
LED Indicators	5V Regulated Power

The power consumption of system components is presented in Table 1. The system is connected to a power supply, which is controlled by a circuit to maintain a consistent voltage for the performance of components like the Arduino. The Arduino acts as a central processing unit, collecting and processing the data from multiple sensors.

The system wirelessly sends the data to an IoT-based cloud server using a Wi-fi module. The data is often updated and stored for real-time monitoring and later analysis. When sensor readings reach a particular point, the system triggers alerts, which will send SMS notifications to local authorities by using the wi-fi module. The areas with limited or no internet connection get alerts through cellular networks. The system architecture shows how all the components are connected to the Arduino.

The ultrasonic sensor is placed near the water bodies to monitor the water level. A combination of an IoT-based technology with a Flood detection system provides instant flood monitoring and alerting. People will receive alerts through Android applications in case of danger. Using IoT technology, we can access the data through this platform to stay informed about flood risks. In case of an emergency, the system will send notifications and alerts to the authorities to prevent flooding.

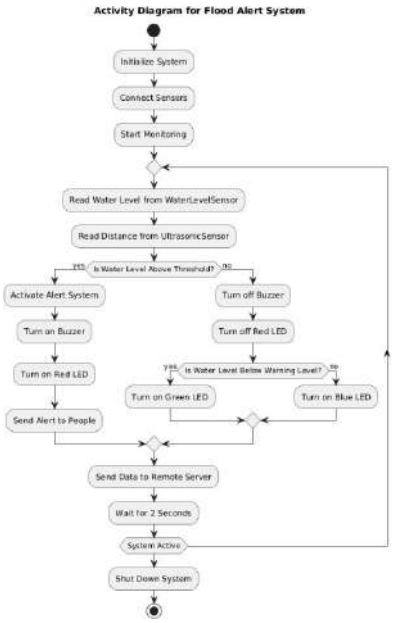

Fig. 2. Activity Diagram

As shown in Fig.2, the activity flow diagram of the flood alerting or deducting system explains the installation of the system components, followed by the connection of sensors for monitoring. After the sensors are connected, the system will start to work and monitor the level of water. The system continuously reads the level of water from the ultrasonic sensor and calculates the distance between the ground level and the water level. Then the system compares the values with a threshold value to find whether there is a flood risk or not.

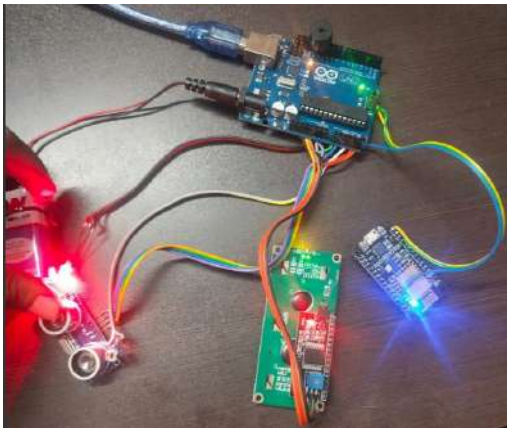

Fig. 3. Red LED indicating danger

Fig.3 shows the alert process started when the water level reaches or crosses the threshold value. The alert system starts to give alerts through the buzzer, LED, and a message. The buzzer starts to make a sound, then the red LED will glow, and an alert message will be received through the Telegram app to the nearby residents.

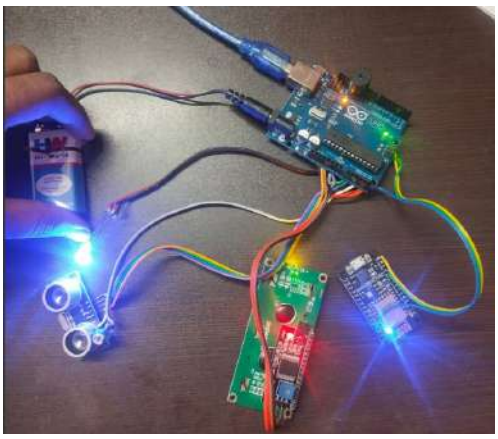

Fig. 4. Blue LED indicating warning

Fig.4 shows that if the LED glows blue, then the government or authorities have to take some precautions so that we can save people from flood risk, and it serves as a warning. Fig.5 illustrates the safe condition, where the green LED is turned on when the water level drops below the predefined risk level.

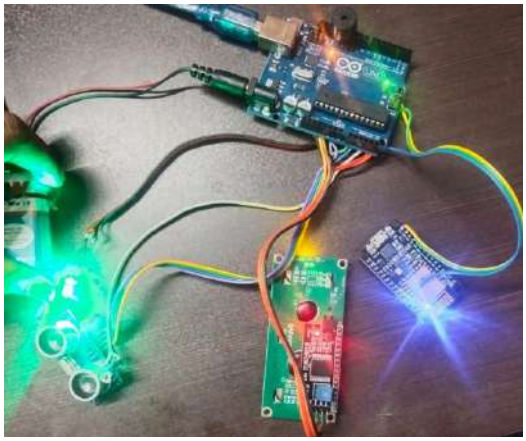

Fig. 5. Green LED indicating Safe

After each measurement, the system transfers the measured data to the Arduino UNO for real-time monitoring. Then the controller continuously monitors the water level and compares it with predefined threshold values. When the water level reaches the threshold value, the controller asks the Wi-fi module to send the alert messages to the people, and the LED indicators start to indicate according to the conditions. The system will continue the mechanism cycle and monitor the environment.

VII. DISCUSSIONS

The purpose of the flood alerting system is to warn people who are in danger. In recent years, floods have caused major consequences, especially in the place 'WAYANAD'.These incidents highlighted the need for the system and discussions on how to alert the people in flood-prone areas. Although natural disasters cannot be managed, but able to take precautions and be ready to take action. Floods have caused many losses, including human, animals, birds, etc...

The objective of this study is to develop a flood alerting system using IoT technology to provide an early warning system. Initially, various approaches for measuring the water levels were analysed through research and studies. Based on this analysis, an ultrasonic water level sensor was selected to measure the water level. The system uses IOT sensors to give a real-time data and a cloud-based server to process the data for real-time monitoring.

Environmental factors such as wind, debris and surface turbulence affect the ultrasonic sensor, which may lead to minor errors due to signal scattering. Additionally, an ultrasonic sensor is fixed in a stable position to reduce vibration-induced noise and ensure consistent distance. To overcome this issue, the system uses:

- Averaging of multiple sensor readings.
- Fixed sensor can reduce vibration.

When multiple flood zone areas trigger the alerts simultaneously, each system operates independently and sends the notifications based on a priority mechanism, ensuring risk zone areas are notified.

Telegram-based notification software was selected due to the immediate, internet-based messaging, group-based, and cost-effective communication compared to traditional SMS services. The idea is to add innovation to the system, which can enhance situational awareness.

The primary objective is to safeguard ourselves from the flood using a flood alerting system and taking some precautions beforehand.

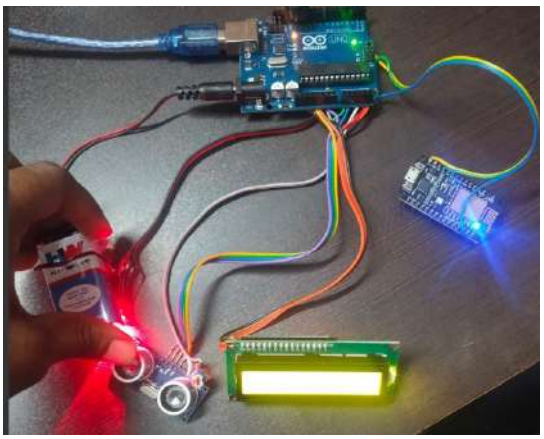

Fig. 6. Final Model

Fig.6 shows the final hardware implementation of the proposed alerting system. The proposed system was functionally tested through a real-time environment and detected changes in water level and sent alerts when the threshold values were exceeded. These results show the system is working efficiently in real-time enablin the telegram notifications.

VIII. CONCLUSION

This paper presented an IoT-based flood alerting system built for real-time water level monitoring and early warning. The system integrates the various components such as an ultrasonic sensor, Arduino microcontroller and Wi-Fi module to detect flood conditions and trigger alerts. The system combines with the IoT technology to provide advanced facilities and is suitable for deployment in flood zone areas. The system successfully detected the water level, triggered an alert when the threshold value was reached and was evaluated through real-time testing under different conditions.

Overall, the proposed flood alerting system is affordable, easy to implement and reliable for real-time monitoring. In the future, the system can be enhanced by integrating additional sensors, GPS, LORA and advanced data analytics to improve accuracy and prediction.

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